

Mendel University in Brno
Faculty of Business and Economy

Lab Exercises for Implementation of Wireless Networks

Bachelor thesis

Supervisor:
Ing. Jiří Balej, Ph.D.

Anna Gurina

Brno 2025

Acknowledgment

I would like to express my deepest gratitude to my supervisor, Ing. Jiří Balej, Ph.D., for his guidance and support throughout the development of this thesis. I am also thankful to those who participated in testing and provided insightful feedback, helping to refine the work. Finally, I extend my appreciation to my family and friends for their unwavering moral support during this project.

Declaration

I confirm by submitting that this thesis entitled **Lab Exercises for Implementation of Wireless Networks** was written and completed by me. I also declare that all the sources and information used to complete the thesis are included in the list of references. I agree that the thesis could be made public in accordance with Article 47b of Act No. 111/1998 Coll., on Higher Education Institutions and on Amendments and Supplements to Some Other Acts (the Higher Education Act), and in accordance with the current Directive on publishing of the final theses.

I am aware that my thesis is written in accordance to Act No. 121/2000 Coll., on Copyright, and therefore Mendel University in Brno has the right to conclude licence agreements on the utilization of the thesis as a school work in accordance with Article 60(1) of the Copyright Act.

Before concluding a licence agreement on utilization of the work by another person, I will request a written statement from the university that the licence agreement is not in contradiction to legitimate interests of the university, and I will also pay a prospective fee to cover the cost incurred in creating the work to the full amount of such costs.

Abstrakt

Gurina Anna. Laboratorní cvičení pro implementaci bezdrátových sítí. Brno: Mendel University in Brno, 2025.

Tato bakalářská práce se zaměřuje na vytvoření série praktických laboratorních cvičení, která se soustředí na implementaci bezdrátových sítí s důrazem na témata odpovídající požadavkům certifikace CCNP ENWLSI. Cvičení se věnují oblastem, jako jsou provozní režimy přístupových bodů, FlexConnect a zabezpečení bezdrátových sítí. Každé cvičení obsahuje stanovený cíl, teoretický základ, popis topologie a vybavení, postup konfigurace a závěrečné shrnutí. Výstupem této práce je sada strukturovaných cvičení, která umožňují praktické procvičení a ověření znalostí v oblasti bezdrátových sítí.

Klíčová slova: Bezdrátové sítě, Cisco, CCNP ENWLSI, Laboratorní cvičení.

Abstract

Gurina Anna. Lab Exercises for Implementation of Wireless Networks. Brno: Mendelova Univerzita v Brně, 2025.

This bachelor thesis is the creation of practical laboratory exercises focused on implementing wireless networks using topics aligned with the CCNP ENWLSI certification objectives. The exercises cover topics such as access point operating modes, FlexConnect, and wireless security. Each lab has a goal, theoretical background, defined topology and equipment, configuration steps, and a summary. The result of this thesis is a set of structured lab exercises that support practical learning of wireless networks.

Key words: Wireless Networks, Cisco, CCNP ENWLSI, Lab Exercises.

Contents

1	Introduction and objective of the thesis	12
1.1	Introduction	12
1.2	Objective	13
1.3	Requirements and Limitations	13
2	Literature Review	14
2.1	Overview of Wireless Networks	14
2.2	Overview of CCNP ENWLSI certification	14
2.3	Research	15
2.4	Books	15
2.5	Documentation and manuals	16
2.6	Final Thesis	17
2.7	Summary of Key Findings	18
3	Methodology and Approach	20
3.1	Student Profile	20
3.2	Selection of Topics and Lab Goals	20
3.3	Review of Available Lab Equipment	21
3.3.1	Cisco Catalyst 9120 Access Point	21
3.3.2	Cisco 3504 Wireless Controllers	22
3.3.3	Other devices	23
3.4	Criteria for Designing and Validating Lab Exercises	24
4	Design of Lab Exercises	25
4.1	Lab Exercise 1	25
4.1.1	Goal of lab	25
4.1.2	Theoretical background	25
4.1.3	Topology and used equipment	29
4.1.4	Configuration Steps and Implementation	29
4.1.5	Summary	42
4.2	Lab Exercise 2	42
4.2.1	Goal of lab	42
4.2.2	Theoretical background	42
4.2.3	Topology and used equipment	44
4.2.4	Configuration Steps and Implementation	45
4.2.5	Summary	53
4.3	Lab Exercise 3	53
4.3.1	Goal of lab	53
4.3.2	Theoretical background	53
4.3.3	Topology and used equipment	56
4.3.4	Configuration Steps and Implementation	56
4.3.5	Summary	64

5	Validation and Feedback	65
5.1	Approach to Validating the Lab Exercises	65
5.2	Results of Student Testing	65
5.2.1	Lab 1	65
5.2.2	Lab 2	66
5.2.3	Lab 3	67
5.3	Summary of Feedback	68
6	Discussion and Conclusion	70
7	References	72

List of Figures

Figure 1: Cisco Catalyst 9120 Access Point (Cisco Systems, 2023a)	22
Figure 2: Cisco 3504 Wireless Controllers (Cisco Systems, 2021a)	23
Figure 3: Lab one exercise topology	29
Figure 4: Cisco Catalyst 9100 Series ports (Benham, 2019)	30
Figure 5: Wireshark capturing a packets from client	38
Figure 6: Cisco Spectrum Expert (Spectrum AQ tab)	39
Figure 7: Cisco Spectrum Expert(Spectrum Spec An tab)	40
Figure 8: CleanAir quality report	41
Figure 9: Captured unclassified rogue APs	41
Figure 10: Lab two exercise topology	44
Figure 11: Clients assignment to VLANs with local profiling	47
Figure 12: Lab three exercise topology	56
Figure 13: FlexConnect AVC application statistics	60

1 Introduction and objective of the thesis

1.1 Introduction

Nowadays, more and more enterprise companies are integrating wireless infrastructure into their networks. As interest in this area grows, so does the demand for skilled professionals who can design, implement, and maintain wireless systems (Coherent Market Insights, 2025). The most widely used manufacturer of such appliances is Cisco. Of course, Cisco is not the only company manufacturing and selling network hardware and software. It also has competitors such as Aruba, Juniper, and Ubiquiti. But Cisco remains the leader in the market for enterprise companies. (Butler et al., 2024)

For working with Cisco equipment, many organisations require that employees have a professional certification as proof of their expertise. In terms of wireless networks, two key concentration exams for achieving the CCNP Enterprise certification are CCNP ENWLSD (300-425)¹ and CCNP ENWLSI (300-430)² (Zippia, 2025) (TechTarget, 2022). Because Cisco has a strong position in the market, many educational institutions have dedicated courses and training programmes that can help students gain knowledge for these certifications (FEE CTU, 2025) (FIT BUT, 2025). At Mendel University, several Cisco courses are available, including a course on Cisco Wireless, but it has not yet been taught. The goal of this work is to contribute to the development of this course by preparing practical laboratory exercises for the CCNP ENWLSI. The exercises will help students gain relevant experience in the configuration and troubleshooting of wireless networks.

It is very important for students to get practical skills in the implementation of wireless networks. Some people find wireless networks to be more challenging than wired ones. Unlike traditional wired networks, where the primary concerns revolve around cable routing, bandwidth allocation, and switch configurations, wireless has more factors that can interrupt the signal that are irrelevant to wired networking. Quite a lot of attention must be paid to the surroundings of the network, so radio signals from other electronic devices do not cause disturbances. Additionally, it must be taken into account that clients can dynamically move. Simply studying wireless protocols and configurations is not enough. This dynamic nature of wireless technology makes the real world experience critical for students. Students need practice to fully understand how to deploy and troubleshoot these types of solution. (Goh et al., 2024)

¹Cisco Certified Network Professional Enterprise Wireless Design – certification that focuses on wireless design concepts, including site surveys, RF design, and wireless architecture (Cisco Systems, 2024c).

²Cisco Certified Network Professional Enterprise Wireless Implementation – certification that focuses on wireless implementation (Cisco Systems, 2024d).

1.2 Objective

The goal of this Bachelor's thesis is to design and implement at least two laboratory exercises for wireless networks that will help students understand and practice the material that corresponds to the main topics of the course "Designing and Implementing Wireless Networks CCNP ENWLSI/I."

Before developing lab scenarios, it is essential to verify that they can be executed within the school lab environment, and it must be ensured that the labs are completed by students within the time range dedicated to the seminar. Upon completion of all necessary steps for implementation, all information will be passed to the lab instructors so they can use them during classes.

1.3 Requirements and Limitations

When designing laboratory exercises, it is essential to consider the resources available at the university. One of the main limitations is the availability of hardware. The university has only a few wireless networking devices, so careful planning is required to maximise their use in different scenarios. In addition, software licenses play a crucial role in determining which features can be utilised in labs. Some tools are available only in limited capacity or with older software versions, which means that exercises must be designed to work within these constraints without requiring additional purchases.(Munawar et al., 2018)

Another important factor is the students level of knowledge. The exercises should add something more, and not just repeat the topics covered in other networking courses. They must be structured in a way that allows students to complete them independently within the allocated seminar time. Furthermore, troubleshooting must be incorporated as an integral part of the exercises to help students develop problem-solving skills in real world networking environments.(Riesen et al., 2019)

2 Literature Review

2.1 Overview of Wireless Networks

Wireless networking has become an essential part of modern IT³ infrastructure. It is widely used both in enterprise and personal environments(Oladele, 2024). In current times, the usage of mobile devices and laptops has drastically increased(Kumar, 2025). Many of them do not even have a suitable port to connect to the wired network. The modern world has made people move constantly. So, the infrastructure needs to adapt to this type of behaviour. The wired network cannot fully satisfy people who are required to change locations multiple times a day. People need to stay connected, always, regardless of where they are. Now it seems like something essential. That is why wireless has become so popular over the years.(Pahlavan et al., 2021)

Despite their differences, there is no need to treat them as opposing technologies. Wired and wireless networks are not mutually exclusive but rather complementary to each other. Although wireless networks offer mobility and convenience, wired networks remain superior in terms of speed, stability, and security. Many enterprise companies use a hybrid networking approach, where wired connections are deployed for critical infrastructure that requires reliability, such as servers, storage systems, and high-performance workstations, while wireless access is provided to employees, guests, and mobile devices. Each situation must be evaluated individually. Enterprise networks require careful planning, which is why it is essential to include a site survey in the process. For some, cabling may be too expensive or, for example, the layout of the building may not allow for the installation of the necessary number of cables. There is no contradiction between wired and wireless networks, only different use cases.(Goh et al., 2024)

2.2 Overview of CCNP ENWLSI certification

To prepare an individual for a specific certification, it is first necessary to understand what this certification entails, what topics are covered, and what the requirements are for passing it.

The CCNP Enterprise Wireless Implementation (ENWLSI 300-430) certification is part of the Cisco Certified Network Professional (CCNP) program. This certification is meant for network engineers who wish to gain knowledge in the implementation and management of Cisco wireless solutions in corporate settings. CCNP ENWLSI demands an advanced understanding of wireless networking that covers security protocols, mobility services, and network automation, unlike entry-level networking certifications. (Cisco Systems, 2024d)

³Information Technology – refers to computer systems and networking hardware and software used for communication over the Internet and other networks(Cisco Systems, 2025l).

According to learningnetwork.cisco.com (Cisco Systems, 2025a), here is a list of topics:

Topic	Weight
FlexConnect	15%
QoS on a Wireless Network	10%
Multicast	10%
Location Services	10%
Advanced Location Services	10%
Security for Wireless Client Connectivity	20%
Monitoring	15%
Device Hardening	10%

2.3 Research

To successfully design exercises, it was essential to look through all available resources. Even though this work is focused on implementation, the design aspect could not be overlooked. Because materials need to represent real world challenges that students can face at their future work. But it would happen in a controlled environment where they can't cause any damage. They will try it out, learn from it, and next time in a situation where their actions will have consequences, they will do everything right. Research must involve analysing academic literature, industry standards, Cisco official documentation, and previous theses that explore topics related to wireless network design, implementation, and troubleshooting. (Henry et al., 2024)

2.4 Books

Books have remained one of the most reliable sources of information over the years. Publishing a book is a complex process. The author is responsible for ensuring the accuracy of the content. To achieve this, a solid research base must be established, and the work must be validated and reviewed by multiple experts before publication. Books usually explore quite a wide topic, but do it from start to finish. (London South Bank University Library, 2025)

One of the most relevant books for us is "CCNP Enterprise Wireless Design ENWLSD 300-425 and Implementation ENWLSI 300-430 Official Cert Guide" by (Henry et al., 2024). This book is widely recognized as a comprehensive reference for individuals preparing to become certified wireless network engineers, as it contains essential theoretical foundations as well as exam-focused technical content. The primary goal of the upcoming course is to prepare students for the exam, which makes this book a key resource for our work. The book covers the same material with which

the example will consist, so it's exploring topics such as: fundamental concepts, best practices, and advanced configurations.

The curriculum has been in preparation for quite some time and was originally intended to be part of the CCNA⁴ curriculum. But was eventually moved to the CCNP⁵ level. A key resource that provides insights into the previous version of this course is the book "Cisco CCNA Wireless" by Tomáš Sochor (Sochor, 2018). This book was originally designed as study material for students taking the Cisco CCNA Wireless course. The book covers fundamental topics of wireless networking, including propagation of radio signals, antenna types, IEEE 802.11⁶ standards, WLAN⁷ security, and network troubleshooting. Since the current CCNP Enterprise Wireless curriculum is based on the latest advancements in wireless technology, some technical aspects have evolved, although foundational concepts in Sochor's book still provide valuable context.

2.5 Documentation and manuals

Official documentation and manuals can also serve as a reliable source of information. Unlike books, which focus on broader topics, documentation is usually focused on a specific subject. The main advantage of such resources is that the vendor or creator of the manual publishes them online, making it easier to keep them up to date. Books, for example, tend to have a problem of becoming outdated quickly after publication. This happens because of the complex process an author faces to publish a book, whereas documentation can be updated much faster. However, at the same time, documentation is not a beginner-friendly product. If someone is just starting, they may find it difficult to read and understand because there is barely any explanation. The author assumes that the reader already has some knowledge in this area and only provides bare information about a certain product or step-by-step configuration instructions. (Lemke, 2022)

For this research, the main source of manuals is www.cisco.com (Cisco Systems, 2025f). Here all necessary information about the configuration of any topology can be found. It is a trustworthy source of information because this website is Cisco's official site and not just a third-party resource, so it can be ensured that the information provided is accurate and up-to-date.

⁴Cisco Certified Network Associate – is an entry-level certification offered by Cisco Systems. It validates a candidate's knowledge and skills in networking fundamentals(Cisco Systems, 2025b).

⁵Cisco Certified Network Professional – is a professional-level certification that covers advanced networking topics(Cisco Systems, 2025c).

⁶IEEE 802.11 is a set of standards developed by the Institute of Electrical and Electronics Engineers (IEEE) that define the protocols for implementing wireless local area network (WLAN) communications in various frequency bands(Cisco Systems, 2008).

⁷Wireless Local Area Network is a network that allows devices to connect and communicate wirelessly within a localized area, such as a home, office, or campus, using radio frequency technology instead of wired connections (Cisco Systems, 2025j).

2.6 Final Thesis

Previous bachelor's and master's theses provide valuable insights into certain topics. Unlike books and official documentation, which present generalized knowledge, final theses often focus on specific case studies. This allows to dive deep into one specific subject. Reviewing previous academic work helps in several ways. First, it provides an understanding of the different approaches taken by other students or researchers. It also offers insight into their experiences and the lessons learned from their work.

The Bachelor's thesis "Implementation of Cisco FlexConnect" by Mikko Jaakonsaari (Jaakonsaari, 2019) was found to be particularly useful as it consists of real-world cases of FlexConnect implementation in various topologies. The study also compares FlexConnect with alternative solutions from other vendors. In addition to configuration, it examines common troubleshooting problems and suggests potential improvements to optimize wireless network performance with Cisco products.

Another important reference is the bachelor's thesis "Preparation of SD-access⁸ laboratory task using Cisco DNA Center⁹" by Michael Varga (Varga, 2022). Although the topic of SD-Access and wireless networking is not directly related, the work can still be applied as it was developed in the same university and with the same lab equipment. The purpose of that work was to create a SD-Access lab exercise with the help of Cisco DNA Center and ISE¹⁰. But serious issues have been faced on the implementation stage of the thesis. The student faced compatibility issues due to old or outdated versions of the software and was not able to obtain fully licensed versions of Cisco ISE or a reliable deployment of DNA Center. Along with this, the hardware and software features that were required to provide full SD-Access capability were not accessible in the laboratory. Consequently, the lab setup was deficient and never really put to the test.

As university infrastructure, available resources, and licensing agreements have not been significantly changed since then, this past experience influenced the present thesis directly. The decision was made to exclude Cisco DNA Center and ISE from the lab scenarios, and to focus on wireless technologies that are fully compatible with the current environment. This avoids repeating known issues and allows the development of stable and verifiable exercises.

A relevant example is the bachelor's thesis "Wireless LAN Deployment: Single-

⁸Software-Defined Access is a Cisco solution that automate and simplify network segmentation, policy enforcement, and device management (Cisco Systems, 2025h).

⁹Cisco Digital Network Architecture Center is a centralized network management platform that provides automation, assurance, and analytics to simplify the management of complex networks (Cisco Systems, 2025e).

¹⁰Identity Services Engine is a security policy management platform that provides secure access to network resources by enforcing compliance and enhancing infrastructure security (Cisco Systems, 2025g).

Sign-On¹¹ with Novell eDirectory¹², Laholms kommun” by Omafume Matthew Enakeyarhe (Enakeyarhe, 2011). The work may be a bit dated, but it remains highly applicable to the current thesis because it was developed using Cisco’s AireOS¹³-based wireless controllers, a platform that is still used in many university laboratories, including the one in this project. Since the core configuration structure and operational logic of AireOS have remained consistent over time, it continues to offer valuable guidance.

The author presents a complete deployment of a wireless LAN in a municipal office environment, using Cisco 5508 WLCs¹⁴ and Cisco Aironet 1142 APs¹⁵. The goal of the thesis was to set up a Single-Sign-On system by connecting Cisco’s wireless network to a Novell eDirectory using a FreeRADIUS^{16,17} server. The work explains how to create a secure wireless network using WPA2-Enterprise¹⁸ and EAP¹⁹ authentication, and how the wireless controller was linked to the external RADIUS server. In addition to basic WLAN and security configuration, the thesis includes the setup of dynamic VLAN²⁰ assignment, certificate installation for EAP authentication, and adjustments on both the WLC and client sides to ensure proper integration. It also documents the troubleshooting process, such as dealing with authentication failures and resolving certificate trust issues. The work illustrates the practical steps needed to build a secure enterprise WLAN and highlights key challenges.

2.7 Summary of Key Findings

The study confirms that the wireless network has become an essential part of today’s infrastructure, driven by the growing use of mobile devices and the need to always stay connected. While wireless networks provide flexibility and mobility, they cannot

¹¹Single Sign-On is a user authentication process that allows a user to access multiple applications or systems with one set of login credentials (Novell, Inc., 2010).

¹²Novell eDirectory is a directory service that stores information about users, devices, and applications on a network (Novell, Inc., 2015).

¹³AireOS is an older operating system used in Cisco wireless controllers, now being replaced by newer technologies.(Magrane, 2021).

¹⁴WLC is a device that manages multiple wireless access points, ensuring consistent settings and policies across a wireless network (Cisco Systems, 2025k).

¹⁵Access point is a device that connects wireless devices to a wired network, extending the network’s coverage (Cisco Systems, 2025i).

¹⁶FreeRADIUS is an open-source software that implements the RADIUS protocol. IT is the most widely used RADIUS server in the world (FreeRADIUS, 2025a).

¹⁷Remote Authentication Dial-In User Service is a networking protocol that provides centralised Authentication, Authorization, and Accounting management for users who connect and use a network service (FreeRADIUS, 2025b).

¹⁸WPA2-Enterprise is a wireless security method that uses a server to authenticate each user individually, providing stronger security for organizations (Edgeworth et al., 2024).

¹⁹EAP is a framework that allows various methods of authenticating users on a network (Edgeworth et al., 2024).

²⁰Virtual Local Area Network is a network segment that isolates traffic within a physical network (Odom et al., 2017).

completely replace wired networks which still outperform them in terms of speed, stability, and security. The enterprise environment often adopts a hybrid networking approach, with wired connections supporting critical infrastructure and wireless solutions providing accessibility to users who require mobility. This highlights the importance of site surveys and careful planning when deploying wireless networks to optimize coverage, reduce interference, and maintain reliability.

To develop an effective lab assignment for CCNP Enterprise Wireless, extensive research was conducted using books, official Cisco documentation, and academic works. The books provide a comprehensive and structured approach to networking concepts, while the official documentation serves as an up-to-date reference for configuration and troubleshooting.

3 Methodology and Approach

3.1 Student Profile

Before starting, it was necessary to understand the expected student profile for this project. The target audience for this course consists of students at Mendel University who have already taken several courses related to networking technologies. The prerequisites for enrolment in the course "EBC-CPEW - Designing and Implementing Wireless Networks CCNP ENWLSD/I" include successful completion of "Computer Networks Protocols," "Switching and Routing in Computer Networks," and "Advanced Routing and Switching CCNA3." These are the same prerequisites as for the course "EBC-CPER - Implementing Routing and Services CCNP ENARSI," indicating that the expected level of networking knowledge for students entering the wireless course is approximately at the CCNA level.

3.2 Selection of Topics and Lab Goals

Before topics for exercises can be selected, it is first necessary to understand what goals must be achieved.

First of all, it is important to consider that this seminar takes place within a university. That means that its content must align with the lecture materials and final exam topics. To ensure this alignment, a consultation was conducted with the person responsible for the course. This step was necessary to understand the specific requirements and expectations for the seminar.

However, this is not the only goal that is intended to be achieved. An additional objective is to help students obtain professional certification, which could be a strong starting point for their careers or help them get higher positions in their field of work. To determine which topics are more likely to appear in the exam, information was gathered from the website learningnetwork.cisco.com (Cisco Systems, 2025a). Based on the provided information, the most frequently asked questions are based on topics: "FlexConnect", "Security for Wireless Client Connectivity" and "Monitoring". Although it is not strictly necessary for laboratory exercises to cover these specific areas, this information can assist in the selection and prioritisation of content when designing laboratory exercises.

It was decided that troubleshooting does not need to be a separate topic but should instead be included as a small part of each exercise. Troubleshooting is an essential part of the configuration in a classroom (Cisco Systems, 2010). It is necessary to acknowledge that mistakes can occur and that students can learn to resolve issues independently. Incorporating troubleshooting into all laboratory assignments ensures that students will develop their problem-solving skills as part of the learning process. This approach also meets the requirement that students should be able to complete their work independently, strengthening their ability to identify and correct problems without direct intervention from the instructor.

3.3 Review of Available Lab Equipment

In cooperation with university instructors, the available resources for implementing wireless laboratory exercises were identified. One of the primary considerations is the physical limitation of hardware in the university environment. The university lab currently has eight Cisco Catalyst 9120 access points and four Cisco 3504 Wireless LAN Controllers. This equipment provides sufficient resources for the development of wireless lab exercises.

The power requirements of the access points are another important factor. Cisco Catalyst 9120 APs require a maximum of 25.5 W of power, but the Cisco Catalyst 3560 switches available in the lab only provide 14.5 W per port (Cisco Systems, 2023b)(Cisco Systems, 2009). This constraint could lead to lessened functionality, performance reduction, or disabling certain features. To avoid such incidents, it is necessary to review the technical specifications of hardware devices. This ensures proper planning of lab activities and helps determine whether students should use specific devices or whether any available equipment in the classroom will suffice.

In addition to hardware, software licensing must also be taken into account. Several tools are available only in limited versions. For instance, DNA Center is limited to version 1, and Cisco ISE is accessible only as a trial. Ekahau Pro²¹ is also available in a restricted format. These limitations have already affected past thesis work at this university, where the attempt to build a lab using Cisco DNA Center and ISE was unsuccessful due to version mismatches and missing licenses. As the hardware and licensing conditions have not changed significantly since that time, it was decided not to use these platforms in the current work. Instead, the focus is placed on implementing wireless lab scenarios that can be reliably tested and demonstrated using supported and stable features of the existing equipment.

3.3.1 Cisco Catalyst 9120 Access Point

The Cisco Catalyst 9120 Series Access Points support up to 400 clients per access point, with 200 clients per radio. The embedded wireless controller feature is not available on this model. The base image of the AP is Cisco IOS XE, version 17.6.4.56. However, upon joining WLC, the AP can detect a different image version on the controller and automatically downloads and runs that image to maintain compatibility. (Cisco Systems, 2023b)

²¹Ekahau Pro – professional planning and site survey tool used for designing and maintaining wireless networks (Henry et al., 2024).

Table 1: Cisco Catalyst 9120 Access Point Specifications (Cisco Systems, 2023b)

Parameter	Value
Model Number	Cisco Catalyst 9120 Series
Processor	Qualcomm quad-core ARM Cortex A53
Memory	2048 MB DRAM, 1024 MB Flash
Storage	Internal storage for firmware and configurations
Network Interfaces	1x 5 Gbps Ethernet (802.3bz Multigigabit), 1x PoE-in (802.3at)
PoE Requirements	25.5W for full functionality, 14.5W for limited operation
Frequency Bands	2.4 GHz (4x4 MIMO), 5 GHz (4x4 MIMO), Optional 6 GHz (Tri-Radio)
Security Features	WPA3, 802.1X, Cisco TrustSec, Encrypted Traffic Analytics



Figure 1: Cisco Catalyst 9120 Access Point (Cisco Systems, 2023a)

3.3.2 Cisco 3504 Wireless Controllers

The Cisco 3504 Wireless Controller is a compact device designed for managing wireless networks in small to medium-sized environments. It provides centralized control of access points and wireless clients, supporting key enterprise features. It supports both centralized and FlexConnect deployments. Overall, this model is mature, but its features are still sufficient for educational environments and practical lab setups. (Cisco Systems, 2021a)

Table 2: Cisco 3504 Wireless Controller Specifications (Cisco Systems, 2021a)

Parameter	Value
Model Number	Cisco 3504 Wireless Controller
Operating System	Cisco AireOS 8.10
Processor	Multicore CPU architecture
Memory	8 GB DDR3 RAM
Storage	32 GB SSD
Network Interfaces	4 x 1 Gbps Ethernet, 1 x RJ-45 Console Port
Max Supported APs	150 Access Points
Max Supported Clients	3,000 Clients
Security Features	WPA3, 802.1X, Rogue AP Detection, Cisco TrustSec



Figure 2: Cisco 3504 Wireless Controllers (Cisco Systems, 2021a)

3.3.3 Other devices

In addition to wireless infrastructure, the school has other network devices. These devices are not the main focus of the thesis but play an essential supporting role.

- The Cisco Catalyst 3560 is used for basic Layer 3 functionality. Although it is a compact switch suitable for lab use, its ports provide limited PoE power, and it does not support multicast routing, which must be considered during lab planning.(Cisco Systems, 2009)
- The Cisco Catalyst 9300 serves as a modern access layer switch, offering full PoE+ support. It provides reliable connectivity for both wireless controllers and access points.(Cisco Systems, 2025d)
- The Cisco Catalyst 2960 is an older switch model used for simple Layer 2 switching tasks. It is suitable for smaller setups and services for network segmentation in design in the labs.(Cisco Systems, 2014a)
- The Cisco 4321 Integrated Services Router is used to provide WAN simulation and Layer 3 routing services. It supports modular interfaces, static routing, and basic network services that are needed for emulating enterprise connectivity.(Cisco Systems, 2024a)

There are other networking equipment in the lab, but the devices listed above are sufficient for the lab scenarios development in this thesis.

3.4 Criteria for Designing and Validating Lab Exercises

For each lab exercise, the objective, individual steps, and methods for verifying the successful completion of each task must be clearly described. Each task must be structured logically, ensuring that the previous steps provide the necessary foundation for the following ones. Each lab should define clear success conditions, specifying the expected output after each configuration step. The duration of the seminar is 2 hours and 30 minutes. The content needs to be clear for the student so that they can finish the exercises independently within the allocated time. Also, it needs to include a part about troubleshooting in case something goes wrong.

After designing the tasks, they must be tested in a lab environment to verify that all configurations work as expected and that the instructions are clear and easy to follow. In addition, a group of students, who are unfamiliar with the material, need to be chosen to try out the exercises. Their feedback will help identify unclear instructions or unexpected challenges, allowing for necessary refinements before the exercises are officially introduced into the course.

4 Design of Lab Exercises

Based on information about student experience with configuring wireless appliances, it was decided that we need to introduce them into topics step-by-step. The smaller topologies and situations need to be practised before moving to more complicated things. You can't learn how to run before you have learned how to walk.

4.1 Lab Exercise 1

4.1.1 Goal of lab

The objective of this lab is to introduce students to wireless problems and to ensure that their practical knowledge reaches the same level as their theoretical understanding. This lab is designed to be simple and to use as little hardware as possible so that students can work in smaller groups to gain a better understanding of the basics. The setup of the WLC and AP will be covered. The logs will be analysed to understand how each application communicates with others on a fundamental level. Students will be introduced to the WLC GUI²² and guided through its basic features. We will try out basic modes for AP, what is available on our hardware, except FlexConnect because we will return to this in the future later. We will also take a look at AP groups and will learn how to make one AP serve multiple SSIDs.

4.1.2 Theoretical background

In the beginning of the exercise, we need to cover theoretical information that is required to execute the practical part. Here are a few essential topics and a brief explanation of what students should remember.

Lightweight AP Connection with WLC

Lightweight access points enable centralized management and simplified configuration. According to Edgeworth (Edgeworth et al., 2024), the process of connection an AP to a WLC occurs in several steps:

1. **WLC Discovery:** The AP detects available controllers using methods such as DHCP²³, DNS²⁴, or local broadcasting.
2. **Establishing a CAPWAP Tunnel:** The CAPWAP tunnel ensures secure communication between the AP and WLC for both control and data traffic.
3. **Downloading Configuration:** The AP downloads the configuration and software from the WLC to maintain network consistency.

²²Graphical User Interface is a visual system that enables users to interact with electronic devices through graphical elements.

²³Dynamic Host Configuration Protocol is a network protocol that automatically assigns IP addresses and other configuration parameters to devices on a network.

²⁴Domain Name System is a system that translates domain names into numerical IP addresses.

4. **Operational State:** The AP starts servicing connected devices and providing network services.

Table 3: Lightweight AP Connection Process (Edgeworth et al., 2024)

Phase	Description
AP Boot	Loading the IOS ²⁵ image, obtaining an IP address
WLC Discovery	Identifying one or more controllers available for connection
CAPWAP²⁶ Tunnel	Attempting to establish a secure tunnel for control messages
Connecting to WLC	Selecting a WLC from the list and sending a connection request
Image Download	WLC informs AP about its software version
Configuration Download	Downloading configuration from WLC and updating existing values
Operational State	AP begins accepting clients
Reset	If the AP is reset by the WLC, it terminates existing clients and CAPWAP tunnels

Cisco AP Operation Modes

Cisco APs can be configured into different modes that determine their behaviour:

- **Local:** The default mode, where the AP serves clients and scans channels for IDS²⁷ events. Off-channel scanning is necessary for Radio Resource Management because it collects information about other channel characteristics, such as noise and interference. In addition, off-channel scanning effectively helps to detect potential security threats. The scanning cycle operates in intervals where the AP scans an alternate channel for 50 milliseconds and then returns to the serving channel for 16 seconds. This scanning cycle repeats continuously, with a default total duration of 180 seconds.(Cisco Systems, 2021c)
- **FlexConnect:** The AP maintains a CAPWAP tunnel to the central WLC but switches traffic locally(Connected mode). In case of WAN failure, the AP

²⁵Internetwork Operating System is Cisco System's proprietary network operating system

²⁶Control and Provisioning of Wireless Access Points is a standardized protocol that enables a WLC to manage multiple APs (Edgeworth et al., 2024).

²⁷An Intrusion Detection System is a security mechanism that monitors network or system activities for malicious actions or policy violations

can continue operating locally (Standalone Mode). This topic will be covered in another lab. (Cisco Systems, 2021d)

- **Monitor:** The AP does not transmit data but functions as a dedicated sensor for IDS events, location-based services, and rogue AP detection. It scans channels at 1.1-second intervals and operates in two modes: the first optimizes tracking via RFID²⁸, while the second supports WIPS²⁹, scanning each channel in 250 milliseconds. (Edgeworth et al., 2024)
- **Sniffer:** In this mode, the Access Point captures all 802.11 wireless traffic on a specific channel and forwards it to a designated machine that has network analyser software, for example, Wireshark. The AP does not offer services to clients at this time. Captured packets include data content, packet size, RSSI³⁰, and timestamps. When the AP is set to Sniffer mode, it is rebooted. After reboot, you can select the band and individual channel that you want the AP to scan. (Edgeworth et al., 2024)
- **Rogue Detector:** In this mode, the AP behaves like a wired sensor because its wireless radio is turned off. It neither transmits nor serves clients but rather looks for ARP³¹ traffic on the wired network for rogue AP and client detection. The AP listens for MAC address details in ARP packets and compares with rogue detection parameters. Upon identifying a match, the WLC alerts an alarm, marking the rogue AP or client as a potential security threat. Another use of Rogue Detector mode is in environments where wireless connection is prohibited. For example, if a company has a no-wireless policy, the AP can look for rogue wireless devices in the network. As soon as any AP is detected in the restricted area, the system will inform the administrator about potential unauthorised wireless use. (Edgeworth et al., 2024)
- **Bridge:** AP functions as a dedicated device to connect two separate network segments wirelessly. In this setup, the AP can create a point-to-point or point-to-multipoint link, basically extending the network cablelessly. It should be mentioned that, in this mode, the AP is utilised exclusively to bridge network segments and does not allow wireless clients. (Edgeworth et al., 2024)
- **Flex+Bridge:** This mode combines FlexConnect and Bridge functionalities, allowing the AP to perform local switching of client traffic while establishing wireless mesh connections with other APs. It enables deployment in

²⁸Radio frequency identification is a wireless technology that uses electromagnetic fields to automatically identify and track tags attached to objects (Henry et al., 2024).

²⁹Wireless Intrusion Prevention System – is a security solution designed to monitor, detect, and prevent unauthorized access and threats in wireless networks (Henry et al., 2024).

³⁰Received Signal Strength Indicator is a measurement used in wireless networking to quantify the power level of the received radio signal at a device's antenna (Henry et al., 2024).

³¹Address Resolution Protocol is a network protocol used to map an IP address to a device's physical MAC address (Henry et al., 2024).

environments requiring both local client connectivity and network bridging across multiple locations. In this mode, the AP can manage client connections and simultaneously bridge different network segments over a wireless back haul.(Edgeworth et al., 2024)

- **SE-Connect:** In Spectrum Expert Connect mode, the AP dedicates itself to spectrum analysis, providing detailed radio frequency (RF) information to external applications like Cisco Spectrum Expert. The AP does not serve clients in this mode but focuses on monitoring the RF environment to identify sources of interference and assist in troubleshooting wireless performance issues.(Edgeworth et al., 2024)

Catalyst 9120 LED Status Indications

Cisco Access Points use LED³² status indicators to provide instantaneous feedback about the operational status of the AP, connectivity, and potential errors. The LED indicators allow network administrators to quickly discern the status of the AP without having to log into the management interface. It is essential to be familiar with these indicators when troubleshooting and providing uninterrupted operation within a wireless network.

Table by Cisco(Cisco Systems, 2023b) provides a view of the different LED states and their interpretations to help understand AP behaviour in all sorts of circumstances such as association status, boot process, software updates, and error conditions.

Message Type	LED State	Message Meaning
Association status	Green	Normal operation, no wireless client associated
	Blue	Normal operation, at least one wireless client associated
Boot loader status	Green	Executing boot loader
Boot loader error	Blinking Green	Boot loader signing verification failure
Operating status	Blinking Blue	Software upgrade in progress
	Alternating Green/Red	Discovery/join process in progress
Access point OS errors	Cycling Red-Off-Green-Off-Blue-Off	General warning; insufficient inline power

Table 4: Cisco Access Point LED Status Indications (Cisco Systems, 2023b)

³²A Light Emitting Diode is a semiconductor device that emits light when an electric current passes through it (Luxes, n.d.).

4.1.3 Topology and used equipment

For implementing this topology, students will require: one Cisco Catalyst 9120 Access Point, one Cisco 3504 Wireless Controller, one L3 switch or router, and a PC. The devices are connected using regular Ethernet cables.

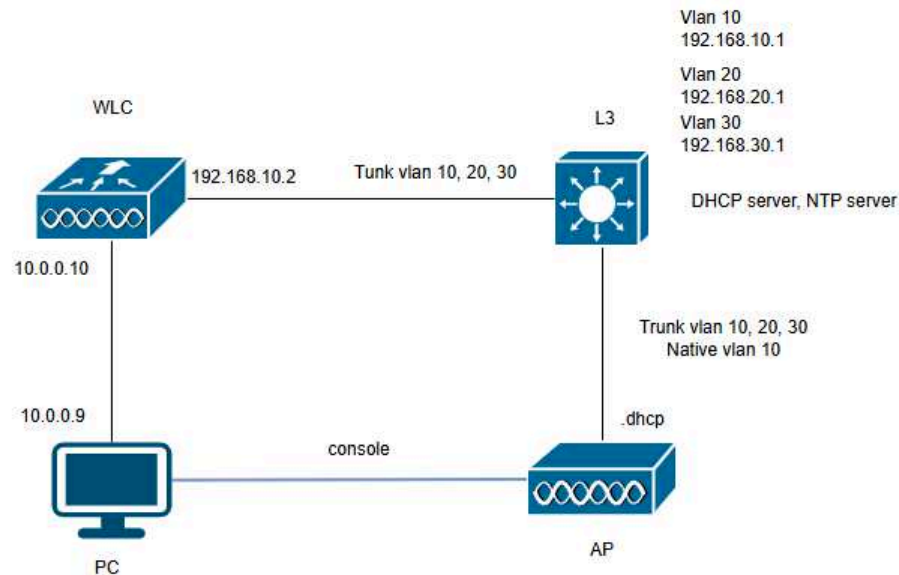


Figure 3: Lab one exercise topology

4.1.4 Configuration Steps and Implementation

In this part of the exercise, students will follow a step-by-step guide to implement the topology described in the previous section.

Step 1: Preparation of the Devices

The teacher needs to divide the students into several groups depending on how many of them are present. Hardware needs to be assigned to each group. To be sure that no previous configuration was applied, students need to perform a hardware reset on each of the appliances.

To perform a reset on a WLC, it is necessary to wait until the appliance finishes the booting process and the configuration dialogue starts. The WLC will ask for a username, and "Recover-Config" must be written to reset the configuration to factory defaults. After that, students will be able to start configuring the WLC. (Cisco Systems, 2013)

Enter User Name (or 'Recover-Config' this one-time only to reset

configuration to factory defaults)

User: Recover-Config

To perform a reset on the AP, the AP must be turned upside down and the reset button located. In figure 4, it is seen that the button is located on the right side. The model on the figure is different from what is available in the class, but it has same button placement.



Figure 4: Cisco Catalyst 9100 Series ports (Benham, 2019)

The AP needs to be powered up and the button needs to be pressed for 20 seconds. It is better to be connected via console cable to the AP to read what processes are going on. If everything is done correctly, the following lines will appear.

```
Button is pressed. Configuration reset activated..  
Keep the button pressed for > 20 seconds for full reset
```

```
Wait for the button to be released ....  
Button pressed for 22 seconds
```

```
Setting env for full reset.Setting env for full reset.
```

Afterward, it is necessary to wait until the process is finished. No configuration needs to be applied to the AP, as it will start to search for the WLC on its own once the booting process is complete.

Then a reset of an L3 switch needs to be performed. These students need to know how to do this from previous classes. After the reset has been completed, students can connect applications according to the topology.

In the end, each device must be successfully reset to factory default. Students need to check the running configuration on each device. If it still has the previous configuration, the step must be repeated one more time.

Step 2: Basic device configuration

At first, the L3 switch needs to be configured. This is something that students are already familiar with, so it will be a good starting point. One of the most important things to establish, though not so obvious at first glance, is the NTP server. Time is a critical factor when trying to establish a connection between the WLC and the AP. If the time is misconfigured, the AP will not join the WLC. In case of a time difference, the following line will appear on the AP.

CAPWAP State: DTLS Teardown

This is why an NTP server is needed.

After that, VLANs and VLAN interfaces must be configured. In this configuration, three VLANs are created on a Layer 3 switch: VLAN 10 (Management), VLAN 20 (Students), and VLAN 30 (Staff). Each VLAN is assigned a corresponding interface with a unique IP address for inter-VLAN routing. Specifically, interface VLAN 10 is configured with IP address 192.168.10.1, VLAN 20 with 192.168.20.1, and VLAN 30 with 192.168.30.1, all using a subnet mask of 255.255.255.0.

In this configuration, two switch ports are trunked to support multiple VLANs. The first port, which is used to connect the Layer 3 switch to another switch, is configured as an IEEE 802.1Q trunk. It is allowed to transport VLANs 10, 20, and 30. This configuration supports inter-switch communication to transport tagged traffic for all relevant VLANs.

The second port, which is bonded to a lightweight access point, also becomes an 802.1Q trunk. It has a native VLAN setup as well. The native VLAN is 10 to be able to recognize untagged management traffic of the AP (for instance, DHCP and CAPWAP discovery packets). The trunk is permitted to transport VLANs 10, 20, and 30 in order to enable the AP to perform both its communication with the wireless controller and client traffic assigned to different VLANs based on SSID configuration. (Odom et al., 2017)

Also, we need to set trunk native VLAN to 10 on an interface leading to AP. In Cisco switching, the native VLAN is used for untagged traffic that arrives on a trunk link. During the boot process, the AP sends untagged DHCP and CAPWAP discovery packets in order to obtain an IP address and locate the Wireless LAN Controller. If the switchport is configured as a trunk, and VLAN 10 is designated as the native VLAN, the switch automatically assigns these untagged packets to VLAN 10 — which is the management VLAN used for controller communication. If the native VLAN were not correctly set, the untagged packets of the AP could be placed on the wrong VLAN (for example, VLAN 1 by default), preventing the AP from receiving an IP address or reaching the WLC and ultimately causing the AP join process to fail.

DHCP servers are also configured on the Layer 3 switch to provide IP addresses to access points and wireless clients. Some address ranges are reserved ahead of time for those devices that will require assigned fixed IP addresses, such as gateways and the Wireless LAN Controller. Three DHCP pools are defined, one per VLAN: VLAN

10 for management, VLAN 20 for students, and VLAN 30 for staff. Each pool specifies the network and default gateway and has DHCP Option 43. It includes the controller's IP address (192.168.10.3) in hexadecimal notation (f104.c0a8.0a03) so that, regardless of which VLAN it belongs to, an access point receives the controller's IP address when it is allotted an IP address. This makes the access point learn and join the controller by CAPWAP tunnelling in a multi-VLAN environment. Although the topology currently has one access point, growth is facilitated in the design through the use of the same Option 43 value in each DHCP pool. (Cisco Systems, 2023d)

After the switching and VLAN infrastructure is set up, the initial configuration of the Cisco Wireless LAN Controller is performed via the console in an interactive setup dialogue. The controller setup process is divided into three logical parts: administrative credentials, IP addressing, and other system parameters.

In the first part of the dialogue, administrative credentials are configured for both the WLC and the access points. These include usernames and passwords for accessing WLC administrative functions, as well as login and enable credentials that the controller will use when managing lightweight APs. (Cisco Systems, 2024b)

```
Enter Administrative User Name (24 characters max): admin
Enter Administrative Password (3 to 127 characters): *****
Re-enter Administrative Password                : *****
Enter User Name for AP (24 characters max): admin
Enter Password for AP (6 to 127 characters): *****
Re-enter Password for AP: *****
Enter Enable Password for AP (6 to 127 characters max): *****
Re-enter Enable Password for AP: *****
```

The second section sets up the network interfaces and IP address of the controller. A static IP address is set up on the service interface, which can be utilised for out-of-band administration or system debugging. The management interface, which is the primary interface utilised for CAPWAP communication with APs and administrative access, is also set here with its IP address, subnet mask, default gateway, VLAN tag, and physical port.

```
Service Interface IP Address Configuration [static] [DHCP]: static
Service Interface IP Address: 10.0.0.10
Service Interface Netmask: 255.255.255.0
```

```
Enable Link Aggregation (LAG) [yes] [NO]: no
mGig Port Max Speed [1000] [2500] [5000]:1000
```

```
Management Interface IP Address: 192.168.10.3
Management Interface Netmask: 255.255.255.0
Management Interface Default Router: 192.168.10.1
```

```
Management Interface VLAN Identifier (0 = untagged): 10
Management Interface Port Num [1 to 5]: 1
Management Interface DHCP Server IP Address: 192.168.10.1
```

During the final phase of the first configuration, certain system-level and wireless-specific parameters are configured to prepare the WLC for operational deployment. This includes the configuration of HA³³, which determines if the controller would run within a redundancy pair. HA is disabled during this configuration because the redundancy problem will be addressed later in the following lessons.

This is then followed by the definition of the virtual gateway IP address. This IP address does not relate to any physical interface, but is used internally by the WLC to support features such as web-based authentication and transparent client mobility between subnets. The mobility group name is also defined, allowing controllers within the same group to exchange mobility and radio frequency (RF) information so that inter-controller roaming and RF coordination can be facilitated.

The configuration dialogue then proceeds with the setting of a default SSID³⁴, StudentLab in this case, which defines the wireless network to be promoted by associated access points. Other parameters include the option to allow the DHCP bridging mode, which is turned off in this configuration. This causes the WLC to act as a DHCP relay rather than bridging DHCP traffic only. The system also accommodates the use of static IP addresses by clients, which is provided to support specific cases where dynamic allocation of IP is not appropriate.

No RADIUS server is implemented here because the current network topology does not include one. However, the system generates an alert, as the default WLAN security policy expects a RADIUS server to provide enterprise authentication. The appropriate country code (CZ) was then defined, which enforced wireless control operation according to local laws and regulations.

The configuration continues with the enabling of the 802.11a, 802.11b, and 802.11g radio networks and the enabling of auto-RF management. The features support the 2.4 GHz and 5 GHz frequency bands and allow the controller to dynamically adjust channel selection and transmit power based on the RF environment.

Finally, NTP³⁵ server is set so as to achieve effective time synchronisation, which is particularly critical for system logging, authentication, and certificate validation. The IPv6 parameters are not established, and as such, the deployment will function exclusively in an IPv4 environment.

With all the configuration parameters fetched and validated, the WLC saves configurations and restarts to effectively complete the initialisation process.

³³High Availability refers to the design and implementation of systems that ensure continuous operational performance and minimal downtime (Henry et al., 2024).

³⁴A Service Set Identifier is the unique name assigned to a wireless local area network (Edgeworth et al., 2024).

³⁵The Network Time Protocol is a networking protocol designed to synchronize the clocks of computer systems (Odom et al., 2017).

```
Enable HA [yes] [NO]: no
Virtual Gateway IP Address: 1.1.1.1
Mobility/RF Group Name: Student
Network Name (SSID): StudentLab
Configure DHCP Bridging Mode [yes] [NO]: no
Allow Static IP Addresses [YES] [no]: yes
Configure a RADIUS Server now? [YES] [no]: no
Warning! The default WLAN security policy requires a RADIUS server.
Please see documentation for more details.
Enter Country Code list (enter 'help' for a list of countries) [US]: CZ

Enable 802.11b Network [YES] [no]: yes
Enable 802.11a Network [YES] [no]: yes
Enable 802.11g Network [YES] [no]: yes
Enable Auto-RF [YES] [no]: yes

Configure a NTP server now? [YES] [no]: yes
Enter the NTP server's IP address: 192.168.10.1
Enter a polling interval between 3600 and 604800 secs: 3600

Would you like to configure IPv6 parameters [YES] [no]: no

Configuration correct?
If yes, system will save it and reset. [yes] [NO]:
```

As a result, each component must be configured as described. Students must verify VLANs, interfaces, trunking, DHCP pools, and WLC settings. If APs do not receive IP addresses or do not join the controller, the time, VLAN, or DHCP settings must be reviewed and corrected.

To verify configuration on a WLC, students can use the following commands: "show interface summary", "show wlan summary", "show clock", "show dhcp summary" (Cisco Systems, 2021b).

Step 3: WLC GUI

Once the WLC has been initialised and rebooted, students can access its web-based interface using a browser. This is typically done by entering the interface IP address into the browser's address bar. It depends on which address needs to be entered because the WLC GUI can be accessed via the management interface or a service port. The login page will prompt for the administrative credentials configured during the initial console-based setup.

After logging in, students are introduced to the WLC GUI, which provides an intuitive environment for managing wireless settings. The first task is to verify the

basic setup and ensure that the controller is reachable, correctly licensed, and that the management interface is operational.

To accept the licence agreement, it is necessary to go to Management → Software Activation → Licenses. Then the available licence must be clicked. After that, the status is set, the End User Licence Agreement is read, and the "I agree" button is pressed. Command "show license summary" can be used to verify success.

If the page does not load or the log-in fails, network connectivity or WLC configuration must be checked.

Step 4: Access Point Join Process

After booting, the AP attempts to obtain an IP address from the DHCP server over the wired interface. This confirms that the AP is physically connected, receives a valid IP address from the management VLAN, and can proceed with the discovery process. The below output is a console log from an access point obtained during the DHCP address assignment process, displaying the network parameters received from the DHCP server.(Cisco Systems, 2023c)

```
ethernet_port wired0, ip 192.168.10.33, netmask 255.255.255.0,  
gw 192.168.10.1, mtu 1500, bcast 192.168.10.255,  
dns1 0.0.0.0, vid 0, static_ip_failover false,  
dhcp_vlan_failover false
```

```
AP IPv4 Address updated from 0.0.0.0 to 192.168.10.33
```

Once an IP address is obtained, the AP attempts to discover the WLC. This typically occurs via DHCP Option 43 or broadcast discovery. Here, the AP successfully retrieves the controller's IP address from Option 43 and receives a discovery response, allowing the process to continue.

```
CAPWAP State: Discovery  
Got WLC address 192.168.10.3 from DHCP.  
Discovery Response from 192.168.10.3
```

Next, the AP initiates a secure DTLS (Datagram Transport Layer Security) session with the controller. DTLS is used by Cisco to secure the CAPWAP control channel between the AP and WLC. This is required for the secure exchange of CAPWAP control messages (Cisco Systems, 2019). Certificate verification must succeed to proceed. If the AP's or WLC's clock is out of sync, this step may fail.

```
CAPWAP State: DTLS Setup  
dtls_verify_server_cert: Controller certificate  
verification successful
```

After DTLS is established, the AP transitions to the Join state, where it formally requests to join the controller. A result code of 0 confirms that the WLC has accepted the join request and will now provision the AP.

```
CAPWAP State: Join
Join Response from 192.168.10.3
AC accepted join request with result code: 0
```

At this stage, the WLC verifies the AP's software version. If the versions mismatch, the AP may download the appropriate image or activate a backup image. This behaviour avoids unnecessary downloads if the correct version is already stored in the AP's backup partition.

```
Version does not match.
Backup image activated. Rebooting AP...
```

After rebooting into the correct image, the AP repeats discovery and DTLS setup and proceeds directly to the join process. This time, since the image matches, the WLC skips the upgrade step.

```
CAPWAP State: Join
Join Response from 192.168.10.3
```

The AP transitions into the Configure state, where it receives configuration details and activates its radios. This stage involves enabling both the 2.4 GHz and 5 GHz radios, setting channels, and preparing to serve clients.

```
CAPWAP State: Configure
Radio [0] Administrative state DISABLED  change to ENABLED
...
[*03/23/2025 13:36:56.3510] Started Radio 1
```

Finally, the AP reaches the Run state and is fully integrated with the WLC. At this point, the AP is ready to transmit SSIDs, serve clients, and forward traffic according to the controller's configuration.

```
CAPWAP State: Run
AP has joined controller Cisco-ac4a.566a.fb02
```

After the process has been finished, it can be verified that AP is present in WLC by clicking on "Wireless" in the menu.

At the end of this step, an AP must successfully receive an IP address, discover the WLC, establish a DTLS session, join the controller, and reach the Run state. If the AP does not join, students must verify DHCP, Option 43, time synchronisation, and trunk settings. The following commands can be used for troubleshooting: "show ap join stats summary all", "show ap summary", "debug capwap client" (Cisco Systems, 2021b).

Step 5: AP groups

This step demonstrates how to configure an access point to broadcast multiple SSIDs using AP groups.

The first task is to create individual interfaces for each VLAN. This is done via the WLC GUI by navigating to: Controller → Interfaces. In the Interfaces tab, a new interface is created by assigning a descriptive name and specifying the corresponding VLAN ID. For each interface, the following parameters must be entered: IP address, subnet mask, default gateway, and the primary DHCP server address (in this case, 192.168.10.1). Once interfaces are created, the corresponding WLANs (SSIDs) must be defined. This is done via: WLANs → Create New. For each WLAN, the students define: Profile Name, SSID, WLAN ID. In this lab, two WLANs are configured: Students (Profile: Students, ID: 20), Staff (Profile: Staff, ID: 30). After creation, each WLAN must be edited by clicking on its ID. The WLAN is then enabled, and the correct interface (corresponding to the VLAN) is selected. In the 'Security' tab, the security type is changed from Enterprise to Personal, which unlocks the field to set a preshared key (PSK). Then a password is configured for each WLAN.

To create an AP group, navigate to WLANs → Advanced → AP Groups → Add Group. A name and optional description are provided for the group. In the newly created AP group, the following configurations are made. In the General tab, the group name and basic parameters are confirmed (an optional NTP server can be defined here). In the WLANs tab, the relevant WLANs are added to the group. In the RF Profile tab, a radio frequency profile can be assigned (this influences power levels, channel plans, and more). In the 802.11u tab, an optional venue type can be defined, which is used in Hotspot 2.0 and advanced client roaming scenarios. The remaining tabs, Location, Ports / Modules and Intelligent Capture, are not used in this basic setup. (Cisco Systems, 2022a)

Finally, in the AP tab, one or more access points are assigned to the AP group. Once added, the AP will reboot and begin broadcasting the defined SSIDs in the group. After reboot, clients should be able to connect to the wireless networks as configured.

If SSIDs are not visible, students must verify interface-to-WLAN mapping and AP group settings. Command for verification: `"show client summary", "show wlan summary", "show ap wlan (802.11-abgn/802.11a/802.11b) <name of AP>"` (Cisco Systems, 2021b).

Step 6: Sniffer mode

To implement sniffer mode within the network topology, an additional access point (AP) must be introduced. An AP operating in sniffer mode is incapable of broadcasting SSIDs or forwarding client traffic. Once the AP associates with the Wireless LAN Controller (WLC), navigate to the WLC's web interface and select Wireless → Access Points → All APs. Click on the desired AP to access its configuration settings. Change the AP mode to sniffer and apply the changes, prompting the AP to reboot. After rebooting the AP, connect a client device to the network. Access

the WLC's CLI³⁶ and execute the following command to retrieve details about the connected client:

```
show client detail <Client MAC address>
```

This command displays details about the connected client, notably the channel in use. Since the AP cannot monitor all channels simultaneously, manual configuration is required.

```
config ap sniff 802.11a enable <channel number>
<IP address of the device where you want to analyze traffic>
<MAC address of AP>
```

After the configuration, launch Wireshark on the designated device. After a brief period, the client packets will be observable. An example of the packet capture display is shown below on a figure 5.

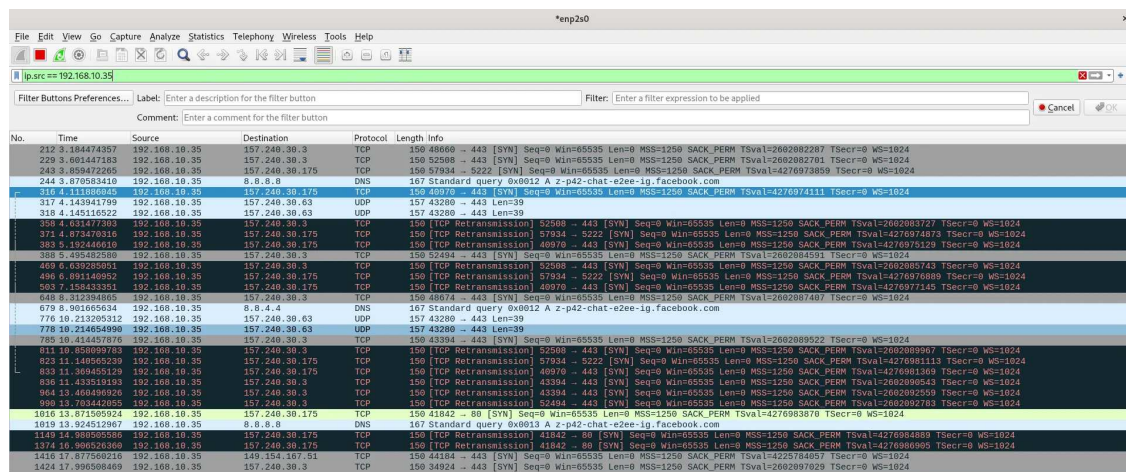


Figure 5: Wireshark capturing a packets from client

If no packets appear in Wireshark, students must verify AP mode, channel configuration, and destination IP address.

Step 7: Se-Connect mode

To activate Spectrum Expert Connect (SE-Connect) mode, access the WLC interface: navigate to Wireless → All APs, select the desired AP, change its mode to "SE-Connect" and apply the changes, prompting the AP to reboot. Ensure that Cisco CleanAir functionality is enabled by selecting Wireless → 802.11a/n/ac/ax and 802.11b/g/n/ax → CleanAir, and verifying that the CleanAir option is active. On a computer equipped with the Cisco Spectrum Expert application (a tool for

³⁶Command-Line Interface is a text-based user interface that enables users to interact with a computer system by typing commands into a console or terminal.

real-time spectrum analysis), open the application and choose to connect to a remote sensor. Enter the AP's IP address and select the appropriate radio band. The application will prompt for a Network Spectrum Interface (NSI) key, obtainable from the WLC under the specific AP's details. After entering the NSI key, establish the connection to utilise the spectrum analysis features. Examples of the Cisco Spectrum Expert interface are shown below on figures 6 and 7. (Cisco Systems, 2014b)

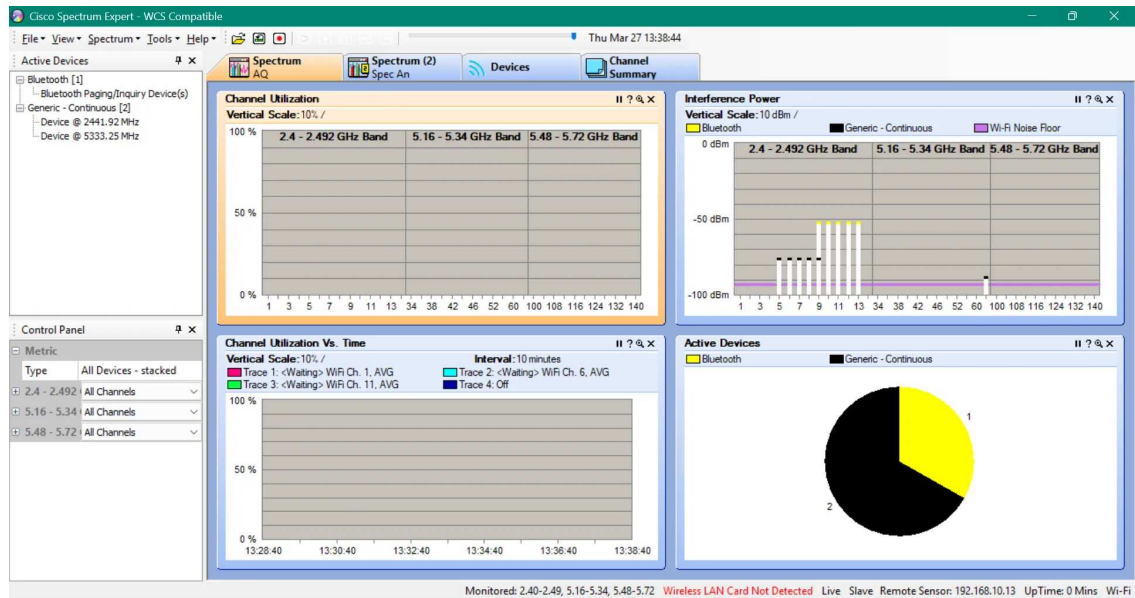


Figure 6: Cisco Spectrum Expert (Spectrum AQ tab)

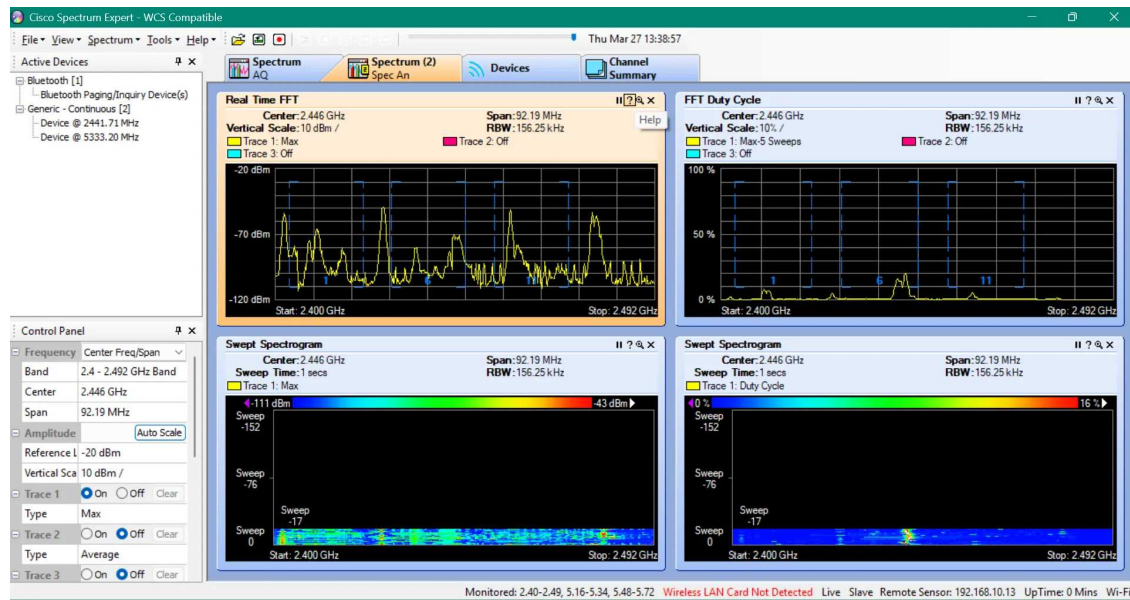


Figure 7: Cisco Spectrum Expert(Spectrum Spec An tab)

If the connection fails, students must verify AP mode, radio band selection, and NSI key accuracy. Or they can try to connect to different channels. Outside of GUI verification, the command "`show <802.11a/802.11b/802.11h> cleanair config`" can be useful (Cisco Systems, 2021b).

Step 8: Monitor mode

In the absence of an Intrusion Detection System, the CleanAir feature on the WLC can be utilised to monitor information collected by access points operating in various modes, including monitor mode. Although CleanAir operates in both monitor and local modes, the scope and efficiency of radio frequency analysis vary between these modes. In local mode, APs provide air quality and interference detection reports for the current operating channel only. In monitor mode, the APs provide reports of air quality and interference detection for all monitored channels.

To configure an AP for monitor mode, navigate to Wireless → All APs, select the desired AP, and change its mode to 'Monitor'. Subsequently, ensure that the CleanAir feature is enabled by going to Wireless → 802.11a/n/ac/ax or 802.11b/g/n/ax → CleanAir, and confirm that the CleanAir option is activated. After a brief period, the Air Quality Report can be accessed under Monitor → Cisco CleanAir. Additionally, information regarding unclassified APs and rogue clients can be viewed in the Monitor → Rogues → Unclassified APs and Rogue Clients. On figure 8, possible output of an air quality report can be found. The figure 8 demonstrates very good signal quality for 2.4 GHz, since there is only one interfering device. On figure 9, students would see rogue APs and SSIDs that they are transmitting.

CleanAir > Worst Air Quality Report (CleanAir only)

802.11a/n Air Quality Report

AP Name	na
Channel Number	na
Minimum Air Quality Index(1 to 100)	na (1-100; 1=worst, 100=best)
Average Air Quality Index(1 to 100)	na (1-100; 1=worst, 100=best)
Interference Device Count	na

802.11b/g/n Air Quality Report

AP Name	AP687D.B4DD.4B68
Channel Number	11
Minimum Air Quality Index(1 to 100)	77 (1-100; 1=worst, 100=best)
Average Air Quality Index(1 to 100)	87 (1-100; 1=worst, 100=best)
Interference Device Count	1

Figure 8: CleanAir quality report

Cisco						
MONITOR WLANs CONTROLLER WIRELESS SECURITY MANAGEMENT COMMANDS HELP						
Monitor						
Summary						
Access Points						
Cisco CleanAir						
802.11a/n/ac/ax						
Interference Devices						
Air Quality Report						
802.11b/g/n/ax						
Interference Devices						
BLE Beacons						
Air Quality Report						
Worst Air-Quality Report						
Statistics						
Controller						
AP Join						
Ports						
RADIUS Servers						
Mobility Statistics						
IPv6 Neighbor Bind						
Counters						
PMIPv6 LNA Statistics						
Preferred Mode						
Optimized Roaming						
CDP						
Rogues						
Friendly APs						
Malicious APs						
Custom APs						
Unclassified APs						
Rogue Clients						
Adhoc Rogues						
Friendly Adhoc						
Malicious Adhoc						
Custom Adhoc						
Unclassified Adhoc						
Rogue AP ignore-list						
Clients						
Sleeping Clients						
Multicast						
Applications						
Lync						
Local Profiling						
Cloud Services						

Unclassified Rogue APs						
Current Filter: None [Change Filter] [Clear Filter]						
Remove Contain Move to Alert						
☐	MAC Address	SSID	Channel	# Detecting Radios	Number of Clients	Status
☐	2a:13:de:7d:da:4d		44	1	0	Alert
☐	34:3a:20:6c:a2:00	faro-free	6	1	0	Alert
☐	34:3a:20:6c:a2:01	eduroam	6	1	0	Alert
☐	34:3a:20:6c:a2:02	mendelu-host	6	1	0	Alert
☐	34:3a:20:6c:a2:10	faro-free	100	1	0	Alert
☐	34:3a:20:6c:a2:11	eduroam	100	1	1	Alert
☐	34:3a:20:6c:a2:12	mendelu-host	100	1	0	Alert
☐	34:3a:20:6c:a2:20	faro-free	1	1	0	Alert
☐	34:3a:20:6c:a2:21	eduroam	1	1	0	Alert
☐	34:3a:20:6c:a2:22	mendelu-host	1	1	0	Alert
☐	34:3a:20:6c:a2:30	faro-free	56	1	0	Alert
☐	34:3a:20:6c:a2:31	eduroam	56	1	5	Alert
☐	34:3a:20:6c:a2:32	mendelu-host	56	1	0	Alert
☐	34:3a:20:6d:8a:b0	faro-free	100	1	0	Alert
☐	34:3a:20:6d:8a:b1	eduroam	100	1	0	Alert
☐	34:3a:20:6d:8a:b2	mendelu-host	100	1	0	Alert
☐	34:3a:20:6f:54:80	faro-free	11	1	0	Alert
☐	34:3a:20:6f:54:81	eduroam	11	1	2	Alert
☐	34:3a:20:6f:54:82	mendelu-host	11	1	0	Alert
☐	34:3a:20:6f:54:90	faro-free	48	1	0	Alert
☐	34:3a:20:6f:54:91	eduroam	48	1	1	Alert
☐	34:3a:20:6f:54:92	mendelu-host	48	1	0	Alert
☐	48:b4:c3:1c:0b:c0	faro-free	11	1	0	Alert
☐	48:b4:c3:1c:0b:c1	eduroam	11	1	0	Alert
☐	48:b4:c3:1c:0b:c2	mendelu-host	11	1	0	Alert

Figure 9: Captured unclassified rogue APs

If data is missing, students must check the AP mode and CleanAir settings.

4.1.5 Summary

This lab showed how to set up a basic wireless network using a WLC and an AP. The students went through the full setup—from preparing VLANs and DHCP, to starting the controller, connecting access points, and creating SSIDs for different user groups.

The main goal of this lab was to help students understand how different parts of a wireless network work together. This setup is the base for all other labs in the course.

By finishing this lab, students practised solving common problems that can occur during the setup. This lab helps students move from learning theory to doing real configuration work. It is an important first step in the preparation for ENWLSI certification.

4.2 Lab Exercise 2

4.2.1 Goal of lab

The goal of this lab is to simulate a simplified BYOD environment by demonstrating how wireless infrastructure can dynamically provide access rights and apply security policies based on the connected device. Students must configure and test local and centralized authentication methods (such as PEAP and web authentication), apply access control mechanisms (ACLs and MAC filtering), and implement profiling-based VLAN assignment.

A FreeRADIUS server is used to provide central authentication and accounting. Students configure user profiles directly on the server, observe authentication traffic, and analyse session logs. In addition, the lab introduces the concept of administrative delegation by creating a Lobby Administrator account.

4.2.2 Theoretical background

Profiling

In networking, profiling refers to the process of identifying and categorising devices connected to a network based on their characteristics, such as the operating system or device type.

AAA (Authentication, Authorisation, Accounting)

Authentication, Authorisation, and Accounting is a framework used to manage and secure user access. It ensures that only authorised users can access network resources, defines what they are allowed to do, and keeps track of their activities.

- Authentication is the process of verifying the identity of a user or device.

- Authorisation determines what actions the user or device is allowed to perform.
- Accounting tracks the consumption of network resources by users. It records details such as session duration, data accessed, and amount of data transferred.

RADIUS Server

RADIUS server (Remote Authentication Dial-In User Service) is a system that helps to manage access in a network. It uses the AAA model to authenticate the identity of users, authorise what they can do, and account for what they have done.(FreeRADIUS, 2025b)

Web Authentication

Web authentication is a mechanism for verifying the identity of a user when accessing a network. When connecting to a network, the user is redirected to a log-in page where they must enter their credentials. After verification, the user is allowed to access the network.(Henry et al., 2024)

EAP Authentication

Extensible Authentication Protocol (EAP) is a framework used to support various authentication methods. It allows different types of credentials, such as passwords, digital certificates, or tokens, to be used to verify user identities. EAP is commonly used in wireless networks and operates over the data link layer, facilitating secure communication between a client (supplier) and an authentication server.(Edgeworth et al., 2024)

PEAP Authentication

Protected Extensible Authentication Protocol (PEAP) is an extension of EAP that enhances security by encapsulating the EAP communication within a secure Transport Layer Security tunnel. This protects the authentication process from potential attacks.(Edgeworth et al., 2024)

BYOD

BYOD (Bring Your Own Device) is a strategy that allows employees to use their personal devices for work-related tasks while connecting to the organization's network. To maintain security and network integrity, BYOD implementations often separate guest users from employees through network segmentation. This separation ensures that employee devices, which may access sensitive corporate data, are isolated from guest devices that typically have limited access to internet services only.(Henry et al., 2024)

MAC filtering

MAC filtering is a method used to authorize network access based on the device's MAC address (Henry et al., 2024).

4.2.3 Topology and used equipment

The required hardware for this lab includes one Cisco 3504 Wireless LAN Controller, two Cisco Catalyst 9120 access points (AP-A and AP-B), one Layer 3 switch for routing and VLAN management, one PC for accessing the WLC GUI, and one external server configured to run RADIUS and web services. To enable internet connectivity, a Layer 2 switch and a router are used. If the lab exercises are performed simultaneously by more than one student, each lab network will connect its own router to the shared Layer 2 switch. This approach allows all the segregated lab environments to access the internet through a shared uplink without compromising the logical segregation of student configurations.

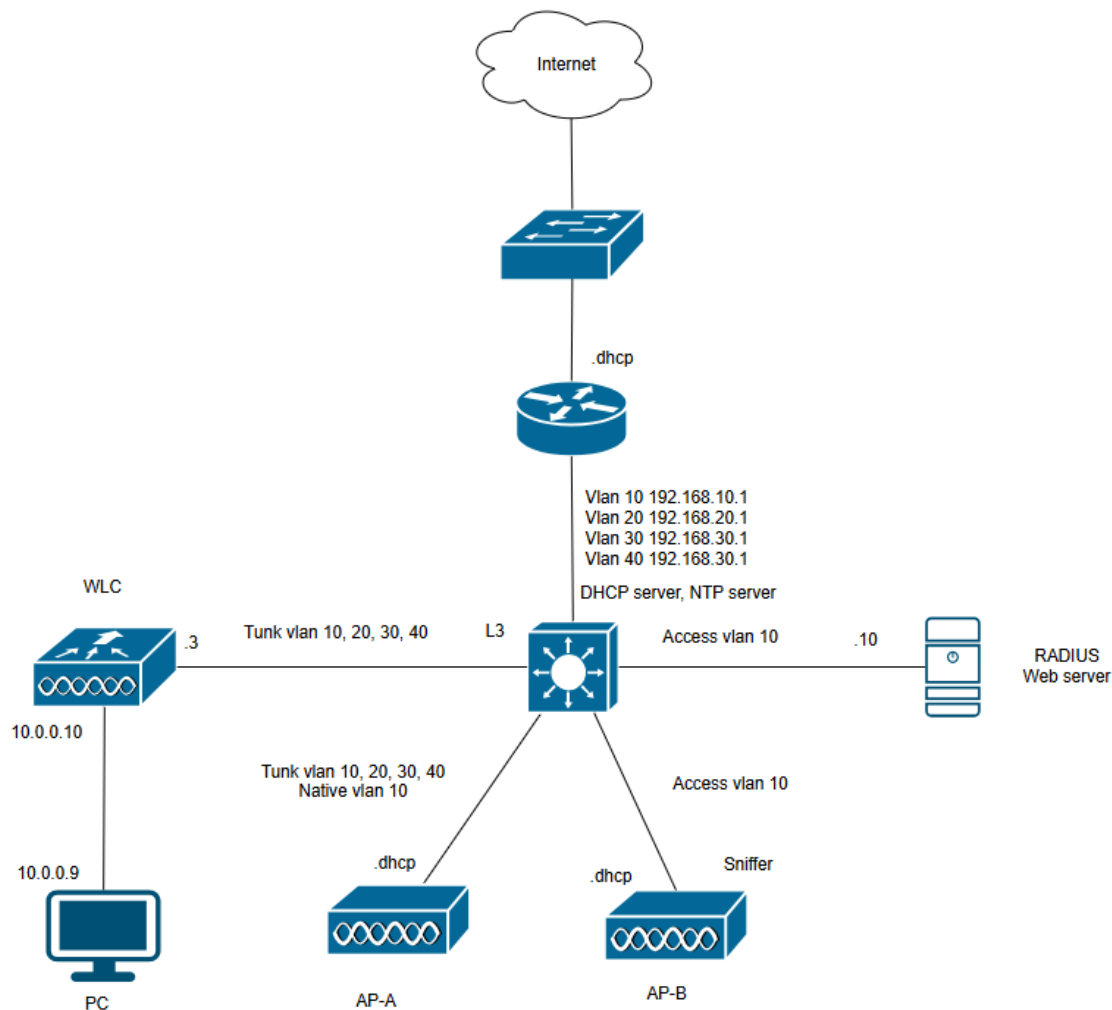


Figure 10: Lab two exercise topology

4.2.4 Configuration Steps and Implementation

Step 1: Initial configuration

Students need to set up devices based on the given network topology. The network design consists of four VLANs, including VLAN 10 for management, VLAN 20 for guests, and VLAN 30 and 40 for employees.

A separate interface needs to be established for each VLAN on the Layer 3 switch and WLC. Correct IP addresses must be assigned to interfaces.

Students must create a group of interfaces that will include VLAN 30 and 40 interfaces. This group enables linking of multiple VLANs in a single WLAN configuration. The interface group must be assigned to WLAN for the proper distribution of VLAN traffic.

Three WLANs must be created. The first one is for trusted users and will use PEAP authentication as a security method. The second will be the guest WLAN with web authentication, and the third is for staff and will have MAC address filtering.

The WLC, AP, and Layer 3 switch need proper trunk configuration to enable tagged VLAN traffic transmission. Students should also not forget to configure trunk native VLAN as in the previous lab.

The Layer 3 switch should function as the DHCP server and the NTP server for all VLANs, and the RADIUS server must be accessible at the end of this step.

AP-B should be in a sniffer mode and should send traffic to the server. This will help students with troubleshooting later.

Step 2: RADIUS authentication

For configuring central authentication with the RADIUS server, students will need to slightly edit the configuration of the server. First, they need to add a user account whose credentials will be authenticated with the RADIUS server to the file located at `/etc/freeradius/3.0/users`. The string for adding a user looks like this `username Cleartext-Password := "your password"`. Then the `client.conf` file in the same directory must be edited to add the WLC as clients so that the server can recognise requests from it. (FreeRADIUS Project, 2018)

```
client WLC {  
    ipaddr = 192.168.10.3  
    secret = your password  
}
```

Next some settings in the `/etc/freeradius/3.0/mods-enabled/eap` file need to be changed. The `default_eap_type` need to be set to `peap`. In the `peap` section `copy_request_to_tunnel` and `use_tunneled_reply` need to be set to `yes`.

After editing the files, the RADIUS server must be started in debug mode using `freeradius -X`. To test if everything was configured correctly, the following command can be used `radtest username yourpassword localhost 0`

PasswordFromWLC.

If an error occurs when starting the server, it may be due to an existing process running on port 1812.

After preparing the RADIUS server, it must be registered on the WLC. This is performed by navigating to Security → AAA → RADIUS → Authentication, clicking "Add New" and providing the IP address, shared secret, and other credentials of the FreeRADIUS server. After that, the server must be linked to the correct WLAN by accessing WLANs → Edit WLAN → Security → AAA Servers and selecting the RADIUS server for authentication. (Henry et al., 2024)

For verification outside of GUI, the following CLI commands can be used: `"show radius auth statistics"`, `"show radius acct statistics"`, `"show radius auth detailed <index>"` (Cisco Systems, 2021b).

Step 3: Security profiles

In this step, students must create a security profile that uses the PEAP method. Here, authentication can be performed locally or centrally. In this part of the lab, the focus is on configuring and testing local authentication.

The configuration process begins with the creation of a local user profile. Students are required to navigate to Security → AAA → Net Users, where a new user account is created. This user profile may be utilized across all WLANs or assigned to a specific WLAN. Additionally, the account can be designated as a guest if needed. In this scenario, the account is associated with a WLAN configured for VLAN 10.

After creating the user profile, a local EAP security profile must be configured. This is done by navigating to Security → Local EAP and creating a new profile, where the PEAP option must be enabled. (Henry et al., 2024)

Once the security profile is ready, it must be associated with the correct WLAN. In the WLAN configuration, students must set the Layer 2 security type to WPA2 + WPA3 and ensure that the security method is set to Enterprise. Then, on the AAA tab, local EAP authentication must be enabled and the previously created security profile must be selected. Additionally, it is important to verify that local authentication is placed at the top of the authentication priority order list to ensure that the WLC uses local authentication first. This configuration ensures that client devices connecting to the WLAN will be authenticated against the locally defined credentials.

Local EAP authentication can be verified through CLI using commands such as `"show local-auth statistics"`, `"show netuser summary"`, and `show "local-auth config"` to confirm user presence, profile setup, and authentication success. Additionally, `"show client summary"` helps verify that clients are connected and authenticated locally, and `"debug aaa local-auth eap method all enable"` can be used for real-time troubleshooting during client connection attempts (Cisco Systems, 2021b).

Step 3: Local profiling

In this step, students are required to configure the local profile on the WLC. The purpose of this task is to show how different types of devices can be automatically assigned to different VLANs based on their characteristics. This allows the network to group clients not just by device type, but by what those types represent in terms of access level or user role.

The configuration starts by navigating to Security → Local Policies. A new policy list must be created and two rules must be added. (Henry et al., 2024)

- All phones (Apple, Android) are assigned to VLAN 30.
- All computers (Windows workstations) are assigned to VLAN 40.

These classifications represent more than just operating systems.

- VLAN 30 represents stuff phones, which should be allowed access to the Internet but isolated from the internal network.
- VLAN 40 represents company-managed laptops or desktops that require access to internal servers.

After creating the list, it must be assigned to the appropriate WLAN. This is done by going to Wireless → WLANs → <WLAN Name> → Policy Mapping and applying the policy list to the WLAN.

Client MAC Addr	IP Address(Ipv4/Ipv6)	AP Name	WLAN Profile	Device Type
56:c4:4e:25:9c:ab	192.168.30.21	APCCDB.935B.649C	StudentLab	Android
96:c1:9a:fa:db:84	192.168.30.22	APCCDB.935B.649C	StudentLab	iPhone 13 Mini
2c:98:11:09:1f:dd	192.168.40.31	APCCDB.935B.649C	StudentLab	Microsoft-Workstation

Figure 11: Clients assignment to VLANs with local profiling

The expected result should match the example in the figure. Each device should be automatically assigned to its respective VLAN. Students can test the configuration by connecting various devices and watching the VLAN assignment.

As a bonus question, students may consider which VLAN would be assigned to a device that does not match any of the defined profiles. This device will be placed into the default VLAN, which students can discover by testing or CLI inspection.

To monitor the results of local profiling, students can go to Monitor → Local Profiling. The WLC provides an overview of the detected device types, shown in a pie chart for easier analysis.

Local profiling can be verified using the command "`show client summary`" to confirm that each connected device is placed in the correct VLAN based on its type (Cisco Systems, 2021b).

Step 4: ACL

To implement traffic control, three ACLs³⁷ must be created. The first ACL is a pre-authentication ACL, designed to block all traffic from users who have not yet completed authentication. The second ACL is intended for guest WLAN and prevents guest clients from accessing internal network resources. The third ACL is for employee mobile devices and blocks access to all local resources except for the internal web server.

It is possible to configure by navigating to Security → Access Control Lists, where each ACL must be created with the appropriate set of rules. Once the ACLs are defined, they must be applied in the correct places in the configuration. There are several methods for applying ACLs: they can be attached to interfaces, WLANs, or local policies. In this setup, the guest WLAN ACL is applied directly to the interface associated with VLAN 20. This is done by navigating to Controller → Interfaces, selecting the interface for VLAN 20, and applying the ACL. The ACL for employee phones is applied through a local policy. This is configured by going to Security → Local Policies, selecting the appropriate policy list, and attaching the ACL. The pre-authentication ACL will be applied later during the web authentication configuration.(Cisco Systems, 2022b)

To validate ACL configurations via CLI, students can use the "`show acl summary`" command to confirm that the ACLs are created and applied correctly. For detailed inspection of each ACL's rules and counters, the "`show acl detailed <acl name>`" command can be used (Cisco Systems, 2021b).

Step 5: MAC filtering

To allow specific devices access to the internal network based on their MAC addresses, MAC filtering must be configured. This is done by navigating to Security → AAA → MAC Filtering on the Wireless LAN Controller. A new entry must be created for each allowed device. Students are required to input the MAC address of the device they intend to connect, assign a profile name, and specify the interface that will be used for the internal WLAN.(Henry et al., 2024)

Step 6: Local Management Users

The Lobby Administrator account is created in Management → Local Management Users. In this menu, a new user is defined with a chosen username and password. The user role is set to LobbyAdmin. This role allows the user to manage guest accounts while restricting access to all other administrative functions of the controller. This configuration is especially useful in environments where guest account creation must be delegated without giving broader configuration access.(Cisco Systems, 2021f)

³⁷ACL is a set of rules used to control network traffic and restrict access to or from network (Odom et al., 2017).

Step 7: Web auth

Students will now configure web authentication to provide controlled access to WLAN users through a web-based login page. The process begins by navigating to Security → Web Auth. The Web authentication type must be left at its default setting, as no custom templates are available for this lab. Students can then configure where users will be redirected after a successful log-in. In this setup, the IP address of an internal web server is entered as the redirection target. In addition, students can choose whether to display the Cisco logo on the login page and have the option to write a headline and a custom message that will appear in the web portal. (Henry et al., 2024)

After preparing the Web authentication settings, students must apply the configuration to a specific WLAN. This is done by accessing the WLANs section and selecting the WLAN where web authentication should be enabled. In the WLAN settings, Layer 2 security must be set to None. In Layer 3 security settings, students must select Web Policy and then choose the specific type of Web Policy.

Several types of web policies are available. Authentication requires users to log in by entering a username and password on the web page. Passthrough redirects users to a web page and grants access without requiring authentication. "Conditional Web Redirect" redirects users based on predefined conditions. Splash Page Web Redirect presents a welcome or terms-of-service page without requiring user credentials. The option On MAC Filter failure triggers web authentication only if MAC address filtering fails.

For this exercise, the option authentication and On MAC Filter failure will be selected. The first is for guest WLAN, the second is for staff WLAN. On a WLAN where MAC filter will be enabled in Security → Layer 2 option, MAC filtering needs to be checked. Also, on the same page, the pre-authentication ACL can be connected to the WLAN.

Web authentication can be performed locally or centrally. For local authentication, the previously created accounts on the WLC are used. For central authentication, the WLC forwards the authentication request to a RADIUS server where the user credentials are stored.

Students are instructed to configure and test both local and central authentication modes to observe the operational differences between them.

Step 6: RADIUS accounting

The purpose of this configuration is to enable user session accounting on the FreeRADIUS server, allowing tracking of session start, updates during the session, and session end, including data usage information.

To prepare the FreeRADIUS server for accounting, it is necessary to edit the `/etc/freeradius/3.0/clients.conf` file. For each client entry, the parameter `require_message_authenticator = yes` must be added. This setting ensures that the server requires the presence of the Message-Authenticator attribute in all RA-

DIUS Access-Request packets, improving security by verifying the integrity of each request.

Next, the `/etc/freeradius/3.0/users` file must be modified. Each user entry must include the parameters `Tunnel-Type`, `Tunnel-Medium-Type`, and `Tunnel-Private-Group-Id`, which instruct the RADIUS server to assign a specific VLAN to the authenticated user. An example configuration is:

```
Tunnel-Type := VLAN,  
Tunnel-Medium-Type := IEEE-802,  
Tunnel-Private-Group-Id := "<VLAN ID>"
```

Following this, it is necessary to configure PEAP as the authentication method. In the file `/etc/freeradius/3.0/mods-enabled/eap`, the parameters `default_eap_type = peap`, `copy_request_to_tunnel = yes`, and `use_tunneled_reply = yes` must be uncommented or modified accordingly.

Attention must also be given to the `/etc/freeradius/3.0/sites-enabled/inner-tunnel` configuration file, ensuring that the `authorize` and `authenticate` sections are uncommented and active.

After preparing the server, it must be added to the WLC. This is performed in the same way as in the previous step where RADIUS authentication was configured. But this time accounting is configured instead. It is also required to enable RADIUS Client Profiling in the Advanced tab to allow additional information to be collected.

The server is then started using the command `sudo freeradius -X`. In parallel, it is recommended to monitor the network interface connected to the WLC using `sudo tcpdump -i enp0s9 -n` to observe live authentication traffic and debug any issues.

Students can now attempt to connect client devices to the WLAN and observe the authentication process on the RADIUS server. A typical session flow captured in the logs appears as follows.

```
21:37:14.883411 In IP 192.168.10.3.59185 > student-VirtualBox.radius:  
**RADIUS, Access-Request (1), id: 0x2d** length: 278
```

This indicates that the client initiated 802.1X EAP authentication by sending an Access-Request packet. This begins the challenge-response handshake.

```
21:37:14.887195 Out IP student-VirtualBox.radius > 192.168.10.3.59185:  
**RADIUS, Access-Challenge (11), id: 0x2d** length: 64
```

The server responds with an Access-Challenge, asking for additional credentials.

```
21:37:14.980166 Out IP student-VirtualBox.radius > 192.168.10.3.59185:  
**RADIUS, Access-Accept (2), id: 0x35** length: 175
```

An Access-Accept response confirms that the client authentication succeeded.

The same process can be followed in another tab. This is an example of what it can look like. The process remains the same. Just here we can get some additional data, for example:

```
(34) User-Name = "student"
(34) EAP-Message = 0x0201000c0173747564656e74
(35) eap_peap: (TLS) PEAP - recv TLS 1.3 Handshake, ClientHello
(35) eap_peap: (TLS) PEAP - send TLS 1.2 Handshake, ServerHello
(35) eap_peap: (TLS) PEAP - send TLS 1.2 Handshake, Certificate
(35) eap_peap: (TLS) PEAP - Handshake state - SSL negotiation finished
successfully
(39) eap_peap: Identity - student
(40) mschap: Found Cleartext-Password, hashing to create NT-Password
(40) mschap: Client is using MS-CHAPv2
(40) eap_mschapv2: MSCHAP Success
(40) eap: EAP session adding &reply:State = 0x9ca92b7e9da13156
(40) eap: Sending EAP Success
```

After a successful connection, students must generate traffic for several minutes to produce session records. These accounting logs are stored in `/var/log/freeradius/radacct/192.168.10.3/detail<number>`.

Students should then review the accounting records, where three types of events can be observed:

A session start record contains fields such as:

```
User-Name = "student"
//The user authenticated as "student".
```

```
NAS-Port = 1
NAS-IP-Address = 192.168.10.3
//This shows which WLC handled the request.
```

```
Framed-IP-Address = 192.168.10.35
//The IP address assigned to the client device during the session
```

```
Airespace-Wlan-Id = 1
//Airespace Wlan ID 1 means this is WLAN Profile 1 on Cisco WLC.
```

```
Acct-Session-Id = "680ab999/a6:d4:f5:2c:65:c4/11"
//The Session ID is unique and used to track all activity for this
session.
```

```
Acct-Status-Type = Start
//This marks the beginning of the accounting session.
```

```
Calling-Station-Id = "a6-d4-f5-2c-65-c4"  
//This is the MAC address of the client.
```

```
Cisco-AVPair = "dhcp-option=\000\014\000\004Anna"  
// Indicates the hostname sent via DHCP was "Anna".  
This is the name of the Android phone.
```

```
Acct-Unique-Session-Id = "0969c92a288105256ee856b510d3088f"  
//Unique session hash generated by FreeRADIUS
```

An interim-update record, sent periodically during the session to report usage statistics:

```
Acct-Status-Type = Interim-Update  
//Confirms that the session is alive and accounting updates  
are working correctly
```

```
Acct-Input-Octets = 7777  
Acct-Output-Octets = 8810  
Acct-Input-Packets = 86  
Acct-Output-Packets = 55  
Acct-Session-Time = 218  
//The user has now transmitted around 7.7 KB  
and received 8.8 KB in 3.6 minutes.
```

```
Cisco-AVPair = "http-tlv=...Mozilla/5.0 ...Android 10 ...Chrome/135.0"  
//The browser was opened on a device.
```

And a session end record, marking the conclusion of the user session:

```
Acct-Status-Type = Stop  
//This confirms the session has ended
```

```
Acct-Terminate-Cause = Idle-Timeout  
//The user session ended because the user was inactive for too long
```

```
Acct-Session-Time = 547  
//This session lasted 547 seconds, or roughly 9 minutes and 7 seconds.
```

```
Acct-Input-Octets = 87721  
Acct-Output-Octets = 14821  
//The user received around 87.7 KB and sent 14.8 KB during the session.
```

By completing this exercise, students gain practical experience in configuring RA-

DIUS accounting, understanding session lifecycle management, and analysing user activity based on network accounting records.

4.2.5 Summary

In this lab, students were able to see how the type of device and the user identity can be used for access control without requiring full BYOD solutions, such as Cisco ISE. They were pushed to link technical actions to policy outcomes and saw how each component of the system must coordinate in order for access control to be successful. This included understanding the relationship between WLANs, VLANs, user profiles, and ACLs. The students also had to work with both internal and external authentication sources, so they needed to configure both the RADIUS server and the WLC correctly and watch the authentication and accounting data flow. This made it clearer for them that network security does not depend on one feature but on consistent policy enforcement across multiple layers of infrastructure.

4.3 Lab Exercise 3

4.3.1 Goal of lab

The objective of this lab is to introduce students to FlexConnect operation and its resilience features, allowing APs to continue to forward traffic during controller outages. Students will learn how to configure APs in FlexConnect mode, create FlexConnect groups, and apply VLAN templates and ACLs. The lab also focuses on advanced resiliency methods such as HA N+N and SSO.

4.3.2 Theoretical background

FlexConnect

FlexConnect is a mode that allows access points to operate in two states, local and FlexConnect mode. In local mode, the AP sends all traffic to the central controller to process it. In FlexConnect mode, the AP can perform switching locally, which allows it to forward traffic locally even when it loses connection to the WLC. The AP caches the information necessary to authenticate and route traffic, allowing operation to continue independently. Traffic in this mode is handled and routed locally by the AP, so it does not have to be sent to the WLC. (Henry et al., 2024)

CAPWAP message aggregation

An advantage of FlexConnect is that it supports CAPWAP message aggregation, where multiple CAPWAP messages can be aggregated into one message. This means that APs and WLCs have reduced overhead in communication, which is particularly beneficial under certain circumstances, such as in a case where the network bandwidth is limited. In AireOS, CAPWAP message aggregation is enabled by default. (Henry et al., 2024)

FlexConnect ACL

The primary difference between FlexConnect ACLs and standard ACLs lies in their application and scope. Standard ACLs are usually configured on the WLC and deal with centrally switched traffic, where the data is forwarded to the controller for processing. In contrast, FlexConnect ACLs are configured on the WLC, with an AP pushing them into enforcement in FlexConnect mode. (Henry et al., 2024)

There can be four types of FlexConnect ACLs:

- **VLAN ACLs:** These ACLs are enforced on the wireless interface of the AP, controlling traffic based on IP addresses or URLs. They help enforce security policies by permitting or denying traffic to destination points. (Henry et al., 2024)
- **WebAuth/Web Passthrough ACLs:** Usually configured for SSIDs that require web authentication or passthrough. They allow client traffic to get to a redirect server before authentication. The routing of traffic is controlled by the AP during authentication. (Henry et al., 2024)
- **WebPolicy ACLs:** Made for scenarios such as conditional web redirection, splash page redirection, and central web authentication. The ACLs specify criteria which shall be used to determine when the redirection of certain web traffic will be triggered, thereby allowing for customised user experiences based on specified circumstances. (Henry et al., 2024)
- **Split tunnelling ACLs:** applied when centrally switched traffic needs to access local resources. This allows some types of traffic to be sent directly to local resources, such as printers or file servers, without crossing the WAN link to the central controller. In this way, bandwidth is used more efficiently. (Henry et al., 2024)

FlexConnect group

FlexConnect Group is a logical grouping of APs that share configuration parameters with WLAN settings, VLAN mappings, and RADIUS server assignments. This makes AP management easier because common configuration settings can be applied consistently to all APs within the same group. An AP that joins a FlexConnect Group inherits the group's configuration settings. (Henry et al., 2024)

Depending on the model of the Cisco Wireless LAN Controller used, each FlexConnect Group supports many APs, with 100 being the theoretical maximum. The FlexConnect Group created by default, commonly called "default-flexgroup", comprises all APs that are not assigned to another group. (Cisco Systems, 2022a)

FlexConnect VLAN template

It is a configuration tool that is used to manage how wireless traffic is assigned to different VLANs when access points operate in FlexConnect mode. This template

allows defining VLAN settings centrally and applying them uniformly to multiple APs grouped together. (Henry et al., 2024)

High Availability. N + N and SSO

High availability (HA) function provides redundancy for WLCs. In FlexConnect, it provides local survivability by allowing the APs to maintain client sessions and even authenticate users independently during controller outages, especially if AAA Survivability features are configured. (Henry et al., 2024)

N + N redundancy: This model involves configuring each AP with a primary and secondary WLC. If the primary controller fails, the AP switches to the secondary controller. During the failover process, the CAPWAP tunnel is re-established, leading to a slight service disruption. This disruption is due to the time it takes the AP to detect the loss, find the backup WLC, perform mutual authentication, and download a new configuration if necessary. The N+N model indicates basic redundancy but is categorised as non-stateful since client-AP sessions are not retained across failovers. (Henry et al., 2024)

Stateful Switchover (SSO): SSO involves a pair of WLCs where one acts as the active controller and the other as a standby. The standby mirrors all the configurations of the active controller. The standby WLC constantly receives updates about the active WLC's client and AP database. In case of failure, it takes over almost immediately into failover, having a minimum impact on client sessions. This setup requires a dedicated redundancy link between the controllers to facilitate state synchronisation. (Henry et al., 2024)

Mobility Group

Mobility Group configuration is enabling seamless client roaming between multiple WLCs. By setting up a Mobility Group, WLCs can share client information, allowing devices to maintain their sessions without interruption when moving across different access points managed by different controllers. This setup is ensuring consistent connectivity and performance in environments with multiple WLCs. (Henry et al., 2024)

LAG

LAG (Link Aggregation Group) is a way to join several network cables into one virtual connection. This gives more total speed and makes the connection more reliable, because if one cable stops working, the others keep the network running. LACP (Link Aggregation Control Protocol) helps set up and manage this joining automatically. In wireless networks, LAG is used to join Ethernet ports together so that the controller has faster and more stable connections. But not all WLCs can use LACP. Some need to have LAG set up manually without any automatic help. The Cisco 3504 WLC does not support LACP. (Henry et al., 2024)

4.3.3 Topology and used equipment

This lab setup uses two Cisco 3504 Wireless LAN Controllers (WLC-A and WLC-B), two Cisco Catalyst 9120 access points (AP-A and AP-B), two Layer 3 switches (L3-A and L3-B), two external servers serving as primary and secondary RADIUS servers, and two standard PCs for accessing the WLC graphical interfaces. All components are connected using Ethernet cables.

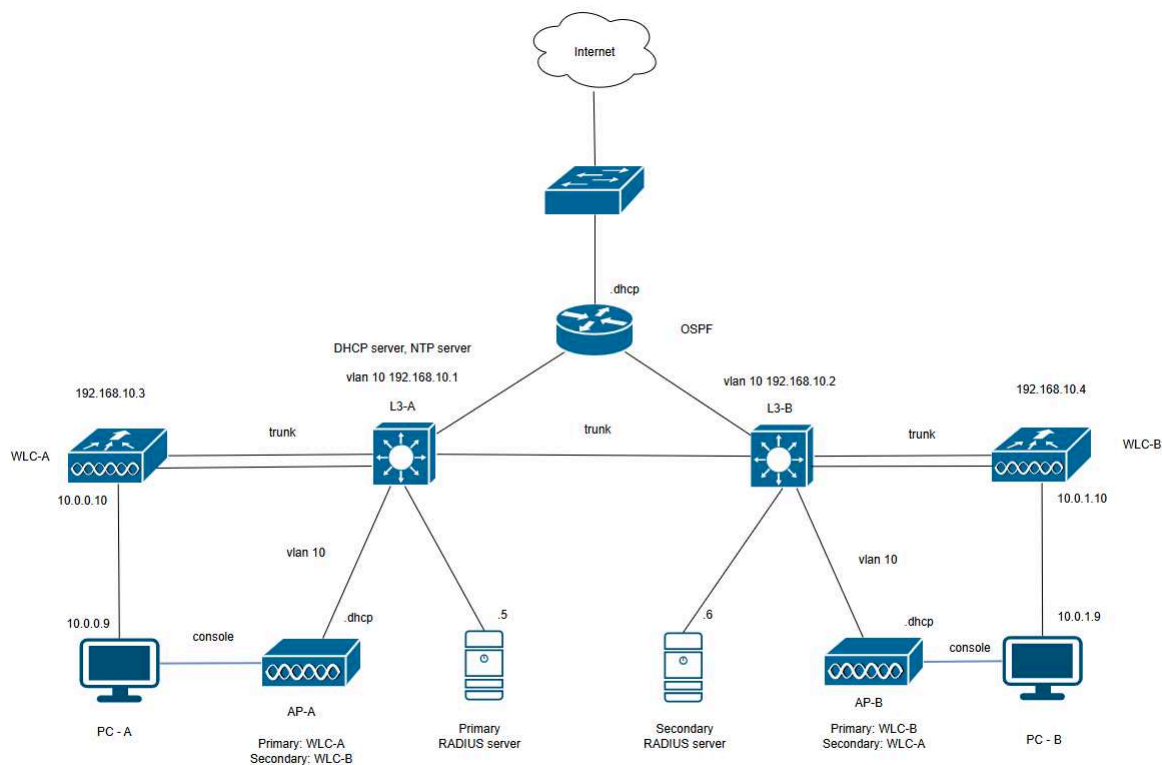


Figure 12: Lab three exercise topology

4.3.4 Configuration Steps and Implementation

Step 1: Initial Configuration

As in the previous labs, all equipment must be reset to factory defaults and the initial configuration performed. Students are required to repeat the basic steps from the previous labs so that the access points can connect to the WLC. In addition, DHCP Option 43 must be configured to include the addresses of both controllers.

For convenience, a WLC-B's graphical interface can be changed to a red colour by navigating to Controller → General → Web Color Theme. Devices, both APs and WLCs, also have to be renamed according to the provided topology. Repeat all the following steps on both WLCs to ensure that the configuration is applied consistently throughout the network.

Step 2: Enabling LAG

To join ports together, LAG mode must be enabled in the general settings of the Wireless LAN Controller. On the other end, a port channel needs to be configured on the switch. After enabling LAG mode on the WLC, the controller will automatically reboot.(Henry et al., 2024)

If the WLC is not accessible after a reboot, students must verify that the switch port channel is configured correctly and that the interfaces connected to it are up.

Step 3: Mobility Group Configuration

The mobility groups are then set up so that controllers can exchange client session information and enable seamless roaming between APs belonging to different controllers. Mobility groups are needed here to ensure session continuity of clients and smooth management in environments with multiple controllers. The step is done through the input of the IP and MAC addresses of the neighbouring WLC, where both controllers will be notified about each other and can manage the client hand-off smoothly. It is achieved through Controller → Mobility Management → Mobility Group → New. Both the IP and the MAC addresses of the neighbouring WLC are required to properly configure the mobility group.(Henry et al., 2024)

For troubleshooting of mobility groups, the following commands can be used: "show mobility summary", "show mobility group member hash" (Cisco Systems, 2021b).

Step 4: FlexConnect VLAN Template Creation

A FlexConnect VLAN template is created to make it easy to configure the VLANs that will be used by FlexConnect access points. This template is needed to specify VLAN mappings for APs in FlexConnect mode and to ensure that traffic can be correctly forwarded and segmented in different VLANs based on the WLAN configuration. The template can be created by navigating to Wireless → FlexConnect VLAN Template. Here, the user selects the required VLAN ID and assigns the corresponding settings to each VLAN. (Henry et al., 2024)

Creation of the template can be validated via CLI with the command "`show flexconnect vlan-name-id summary`" (Cisco Systems, 2021b).

Step 5: Creating a FlexConnect Group

Once the VLAN template is created, a FlexConnect group must be established by going to Wireless → FlexConnect Groups → Create. After that, select the VLAN template created earlier to associate the template with the group. This association ensures that the APs within the group use the same VLAN configurations.(Henry et al., 2024)

When you switch the APs to FlexConnect mode, you must assign them to the FlexConnect group. The APs will then inherit group settings, ensuring that any configuration changes will be uniformly applied across all group members.

Next, navigate to the WLAN VLAN Mapping tab. The purpose of this step is to map each WLAN to its appropriate VLAN to ensure that client traffic flows to the intended network. WLAN ID 10 is mapped to VLAN 10, which is the management VLAN in this topology.

The following commands can be used for validation and troubleshooting: "show flexconnect group detail <Group Name>", "show flexconnect group summary", "show ap flexconnect module-vlan <AP name>" (Cisco Systems, 2021b).

Step 6: WLAN Configuration and Enabling FlexConnect

In the WLAN configuration, select the desired WLAN and open the Advanced tab. From here, enable FlexConnect local switching and learn local client IP address. This ensures that client data will be locally switched at the AP in FlexConnect mode, reducing reliance on the WLC and improving performance.(Henry et al., 2024)

Step 7: Enabling FlexConnect Mode on AP

After enabling FlexConnect mode on the AP, go to Wireless → All APs and select the appropriate AP by name. Change the AP mode to FlexConnect. Once FlexConnect mode is activated, the AP will not reboot. It will continue to function normally. The FlexConnect tab will appear, which will allow for further configuration.(Henry et al., 2024)

For one of the APs, settings will be configured manually even though it is part of the FlexConnect group. Shows the flexibility of FlexConnect mode, where individual APs can still be configured separately for specific requirements or troubleshooting, even when they are part of a larger group with shared settings. In the FlexConnect tab, enable VLAN support, set the native VLAN to 10, and map the WLAN ID to the appropriate VLAN. Ensure that the same configurations are applied to both WLCs to maintain consistency across the network.

Step 8: RADIUS servers

The first part of this step follows the same procedure as in the previous lab. The task is to set up two RADIUS servers and to have them registered on the Wireless LAN Controller. By now, the students should be familiar with how to configure RADIUS, since it has already been done previously.

Additionally, in the WLAN configuration under the Security tab, it is necessary to activate the RADIUS Server Override Interface, enable RADIUS Authentication Survivability, and Local EAP Authentication, and ensure that RADIUS is placed at the top of the Authentication Priority Order list. In the Advanced tab, the option Allow AAA Override must also be enabled.

Subsequently, within the FlexConnect Group settings, both RADIUS servers need to be added again, with the appropriate server type specified. Finally, in the

section Security → AAA → RADIUS → Fallback, the Fallback Mode should be set to Active. (Henry et al., 2024)

Step 9: Connecting to the internet

Now the teacher needs to configure a gateway on the switch to connect to the school network, and then the students will all connect to the switch to have access to it.

Step 10: Testing the Wireless Connection

To test the wireless connection, connect an external device to the network. This device will simulate a user device that connects to the wireless network. Ensure that the device connects successfully to the configured WLAN and obtains an IP address via DHCP. Once connected, test the functionality of the network.

Step 11: FlexConnect ACL

To configure a FlexConnect ACL, it is necessary to navigate to Wireless → FlexConnect ACL and press Create New. Let us imagine a situation where there is one WLAN used by both students and network devices. For example, this network is dedicated to a class where students are writing exams, so internet access should be restricted. (Henry et al., 2024)

For this, a new ACL is created with the name NoInternet, and two rules are added in this sequence: the first rule allows communication within the local network for the second part of the subnet, and the second rule denies all other traffic. After creating the ACL, it must be applied by going to the FlexConnect Group, opening the ACL Mapping tab, and applying the ACL in the egress direction on VLAN 10.

If clients still reach blocked destinations, students must verify the order of the rules and the direction of the ACL. The following commands can be helpful for it: `"show flexconnect acl summary"`, `"show flexconnect acl detail <ACL name>"`, `"debug client <MAC address>"` (Cisco Systems, 2021b). Or you can restart an AP to force it to download a new image from WLC.

Step 12: FlexConnect AVC

To configure AVC for FlexConnect, a profile must first be created. This is done by going to Wireless → Application Visibility and Control. A new profile can then be created, and rules can be added to the profile. As an example, the rule will define to drop all HTTP traffic. The profile must be applied to the FlexConnect group after it is created. This profile is then applied in the WLAN AVC Mapping tab by selecting the WLAN ID, enabling Application Visibility, and assigning the Flex AVC Profile. The Application Visibility field has three options: WLAN Specific (different profiles per WLAN), Enable (activates AVC), and Disable (deactivates AVC). (Cisco Systems, 2018)

The WLAN selected must be edited after mapping. In the WLANs section, select the target FlexConnect WLAN and enable the Application Visibility option within the QoS tab.

To verify the configuration, students need to generate application traffic. The WLC will display in-depth statistics on application visibility and traffic classification after about 90 seconds. Also, students can try to perform the command "curl -I" to a web page that is using http, and as a result, they must not receive a response. In figure 13, an example of AVC statistics can be seen.

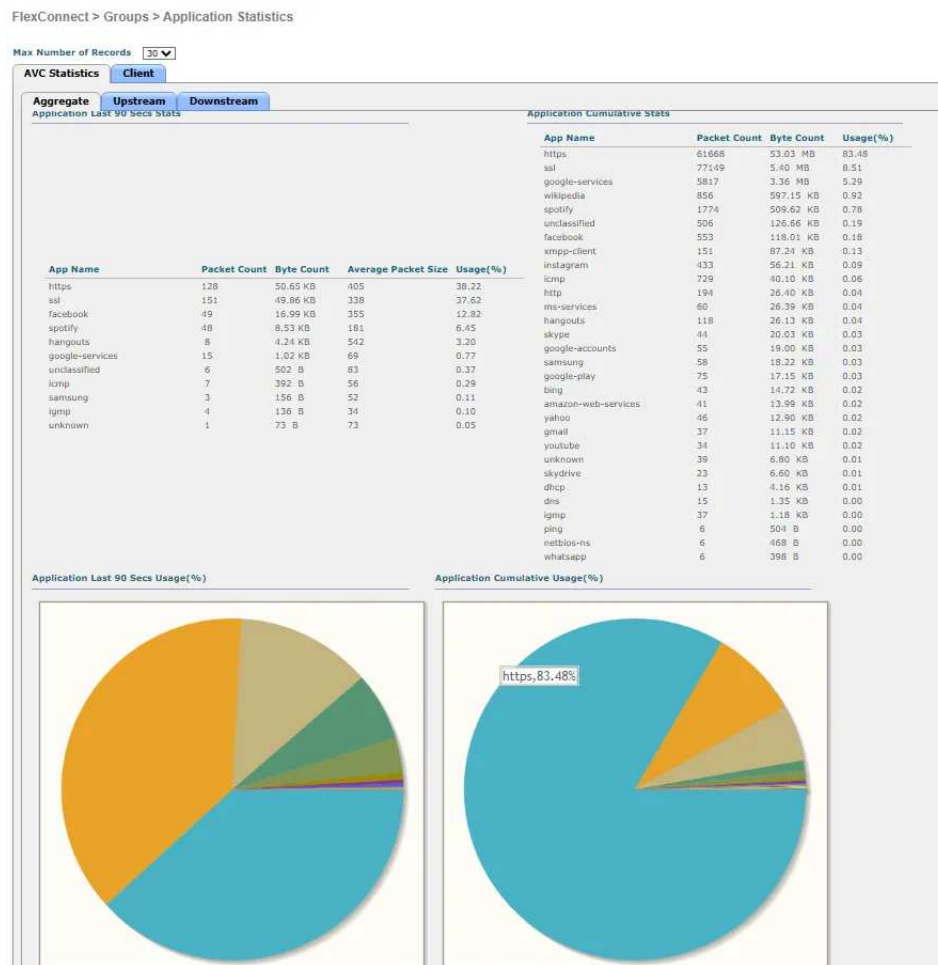


Figure 13: FlexConnect AVC application statistics

At the end of this step, the AVC profile must be visible under the FlexConnect group and mapped to the correct WLAN. If traffic is not classified or HTTP is not blocked, students must verify the AVC status and traffic generation. Commands: "show avc statistics client <MAC address>", "debug avc" (Cisco Systems, 2021b).

Step 13: FlexConnect Resiliency Scenarios N + N

For HA N+N, it is necessary to configure each AP individually, because, without an external device, AireOS does not support centralised HA configurations. Therefore, every access point must be manually configured for redundancy with both controllers. To configure HA, navigate to Wireless > All APs, and select the AP of interest. In the High Availability section, configure the names along with IP addresses of both primary and secondary WLCs. For example, in WLC-A for AP-A, the primary WLC will be WLC-A, and the secondary WLC will be WLC-B. Those actions need to be repeated on each AP individually.

Ensure that AP fallback is enabled in the Controller > General. AP fallback is a function that allows access points to reconnect to their primary WLC once it is up again after the failure.

After HA is configured, disconnect one and then both WLCs to observe network behaviour.

In case one WLC is disconnected with HA configured, APs will detect the loss of their primary WLC and automatically attempt to connect to the secondary WLC. This failover process is designed to occur automatically and in a short time window. However, in cases where WLC configurations are not identical, the AP may be required to download new settings, which can result in a brief service disruption or even a reboot. Furthermore, during the failover interval, the AP transitions to the standalone mode, continuing to serve previously associated clients via local switching. New client associations, however, may be delayed until the CAPWAP connection with the new controller is successfully re-established. Because DHCP Option 43 is already configured on the switch, the AP can discover the alternate WLC more efficiently. This results in a faster reconnection process compared to setups where no DHCP-based controller discovery is available.

The log below, captured from the AP, shows that the AP attempts to reach the WLC five times. After the maximum number of retransmissions is exceeded, the AP enters standalone mode to maintain client service.

```
Re-Tx Count=1, Max Re-Tx Value=5, SendSeqNum=33, NumofPendingMsgs=1
Re-Tx Count=2, Max Re-Tx Value=5, SendSeqNum=33, NumofPendingMsgs=1
Re-Tx Count=3, Max Re-Tx Value=5, SendSeqNum=33, NumofPendingMsgs=1
Re-Tx Count=4, Max Re-Tx Value=5, SendSeqNum=33, NumofPendingMsgs=1
Re-Tx Count=5, Max Re-Tx Value=5, SendSeqNum=37, NumofPendingMsgs=5
```

Max retransmission count exceeded, going back to DISCOVER mode.

```
Dropping msg CAPWAP_WTP_EVENT_REQUEST, type = 34, len = 505,
eleLen = 513, sendSeqNum = 37
```

```
...Vendor SubType: AP_CDP_CACHE_PAYLOAD(24) len: 501 vendId 409600
```

```
Dropping msg CAPWAP_WTP_EVENT_REQUEST, type = 34, len = 863,
eleLen = 871, sendSeqNum = 37
```

```
Dropping msg CAPWAP_WTP_EVENT_REQUEST, type = 34, len = 38,
```

```

eleLen = 46, sendSeqNum = 37
....TLV: TLV_ECHOPAYLOAD(42), level: 0, seq: 0, nested: true
Dropping msg CAPWAP_WTP_EVENT_REQUEST, type = 34, len = 64,
eleLen = 72, sendSeqNum = 37
....TLV: TLV_APETHERINTFPAYLOAD(46), level: 0, seq: 0, nested: true
Flexconnect Switching to Standalone Mode!

```

To verify the current switching mode and client connectivity, the AP CLI can be used to check if the clients remained connected during the standalone period. The command output below shows that one client is actively using local switching, while another is not yet active.

```

AP-B#show flexconnect client counter
Flexconnect Clients:
      client centrally-switched-counter locally-switched-counter
56:C4:4E:25:9C:AB                0                428
68:7D:B4:DF:B2:E0                0                0

```

If both WLCs are disconnected from the network, the AP will continue to operate in a standalone mode and serve all clients using locally switched traffic. Once one or both controllers are restored, the APs will use the fallback mechanism to reconnect to their assigned primary controllers.

If some of the functions didn't work as planned, those commands can be used to find an issue: "show ap config general <AP name>", "show flexconnect client counter", "debug capwap events enable" (Cisco Systems, 2021b).

Step 14: FlexConnect resilience scenarios SSO

To configure SSO, it is necessary to add an additional connection link between the two WLCs. This is required because the peering process takes place between the redundancy management interfaces. If the WLCs are not connected, the peering process will not occur.(Cisco Systems, 2021e)

After the physical connection is made, IP addresses must be assigned to the redundancy management interfaces. Then, navigate to Controller → Redundancy. Within this menu, the roles of the WLCs are defined: one is selected as the primary, and the other as the secondary. The IP address of the peer WLC's redundancy management interface of the peer must be specified and SSO must be enabled. Once the configuration is applied, the controller reboots and initiates the peering process.(Cisco Systems, 2021e)

Successful initiation of peering and role negotiation is indicated by the following log messages:

```

Starting Redundancy: Starting Peer Search Timer of 120 seconds
Initiate Role Negotiation Message to peer
Found the Peer. Starting Role Determination...ok

```

After, the log will display the role of the local and peer WLCs, confirming that the SSO pairing has been completed:

```
Redundancy Mode SSO enabled
Local unit is ACTIVE (or STANDBY HOT)
Peer unit is STANDBY HOT (or ACTIVE)
```

When the WLC is in STANDBY HOT mode, it continuously synchronizes its configuration with the active controller. The graphical user interface is not accessible on the standby controller in order to prevent configuration conflicts.

To validate the failover scenario, the active WLC is manually disconnected. In response, the disconnected WLC enters maintenance mode, as indicated by the following output:

```
Entering maintenance mode as all the Local Ports are down
on Standby WLC
```

```
Gw Reachability Counters.....
-----
Gw pings succesfully sent.....: 60
Gw pings failed to send.....: 0
Gw responses received.....: 0
Current consecutive Gw responses lost.....: 60
High water mark of Gw responses lost.....: 60
-----
```

```
Entering maintenance mode...
```

The standby WLC completes the switchover and assumes the active role. The log confirms this transition with the following entry:

```
HA completed successfully, WLC switch over detection time : ~1360 msec
and APs switch over time : ~1 msec
```

From the access point's perspective, no failover is detected. The AP continues operation without entering standalone mode, nor does it initiate a new WLC discovery process as seen in N+N redundancy scenarios. Client traffic remains uninterrupted throughout the transition.

The following commands can be used for troubleshooting: "show redundancy summary", "debug redundancy infra", "debug redundancy keepalive" (Cisco Systems, 2021e).

Step 14: FlexConnect resilience scenarios RADIUS

Students can test how the system reacts when the primary RADIUS server goes offline. After the primary server is disconnected, new users should connect through

the backup server. When the primary server is available again, the controller should automatically start using it for new authentication requests.

It is important to understand that the backup RADIUS server responds more slowly. Modern phones have a short timeout and might stop trying before the server replies. In that case, the phone may say it cannot connect, even though the server will send a reply a bit later and finally the client would be able to connect. Students should make sure the phone is not switching to another Wi-Fi network during this time and just wait a little longer. The reply will come, but with the delay.

4.3.5 Summary

By completing this lab, students saw that resilience in wireless networks is not guaranteed by a single feature. FlexConnect switching alone is not enough without planning for controller redundancy. The function relies on correct communication between the controller and the access point was shown. When this link breaks, the AP must continue to serve clients, but not all services behave in the same way. The students also observed how failover time and configuration mismatches affect the user experience. In SSO, this handoff is invisible. In N+N, it is often not. The lab showed that failure tests are as important as normal tests.

5 Validation and Feedback

5.1 Approach to Validating the Lab Exercises

The lab exercises were tested by the author and two students. A larger group was not involved due to the limited access to equipment and time restrictions of the participants. The main goal was to closely observe the process and to receive direct feedback after each step. Testing with more people would have required more resources and would have made detailed supervision more difficult.

The two students who participated were chosen because they match the expected profile of the lab's target users. Both are students of Mendel University and are enrolled in the same study programme. They have already completed other courses covering networking topics. This implies that they had sufficient prerequisites to test the lab as intended. One of the students was familiar with wireless networking only on a theoretical level and had not done any configuration before. The other student had general networking experience, but not specifically with wireless controllers.

The author of the lab acted as the test facilitator. During the testing sessions, the author observed the students, answered clarification questions when needed, and noted where they encountered difficulties. This approach made it possible to directly adjust the instructions and ensure that the lab could be completed independently in a future classroom setting.

The testing was done using real hardware in a lab environment, with both students following the same steps, but working separately. Measurements of how much time each step required were taken, and any system delays (such as AP reboot time) were also recorded. The goal of the validation was not only to confirm technical correctness but also to check whether the instructions were understandable and followed the logic of what had been taught in previous classes.

5.2 Results of Student Testing

5.2.1 Lab 1

The original version of the first lab included visual instructions with step-by-step images. However, during the test, the facilitator observed that students were seen following the instructions without actually understanding them. There was no deep engagement with the task or the effort to understand the configuration logic. The steps were too exact, so the students could complete them without thinking about what they were doing.

As a result, the lab was redesigned. Instead of giving complete visual directions, the instructions were replaced by text, divided into two logical sections. The first part contained short and open sentences that gave brief instructions. The second section contained more expansive help for students, which they can open if they want further assistance or to check if they finished task correctly. This new format

required students to read, understand, and apply their knowledge instead of copying the instructions.

This change improved the learning process. The students asked more questions and critically thought about each step. At the end of the session, both participants showed an understanding of the topic and their actions during the lab. Their understanding of the lab objective expanded, making them more confident in repeating the configuration without additional assistance. The students were also able to implement each configuration step correctly without critical errors.

Here is a breakdown with step-by-step time stamp of how long it took to complete each of the tasks.

Step	Experienced student (min)	Inexperienced student (min)
Step 1: Preparation of the Devices	15	23
Step 2: Basic Device Configuration	35	55
Step 3: WLC GUI	7	12
Step 4: Access Point Join Process	17	30
Step 5: AP Groups	23	33
Step 6: Sniffer Mode	20	27
Step 7: SE-Connect Mode	19	26
Step 8: Monitor Mode	15	21
Total Time	151	227

Both students completed the lab successfully and were able to explain the role and function of each step. The results indicate that the lab is accessible to students with varying levels of prior knowledge.

5.2.2 Lab 2

The students gave helpful feedback during testing. Originally, Lab 2 and Lab 3 were in a different order. After completing both labs, the students suggested switching their order. They noticed that some parts of Lab 3 were easier to understand after doing Lab 2, especially the configuration of interface groups and the use of RADIUS. They also pointed out that Lab 3 was longer and more difficult, so it made sense to do Lab 2 first to balance the workload between them. This suggestion was accepted, and the labs were rearranged.

Students found the RADIUS accounting part of the lab especially useful. The topic had been covered a lot in theory during lectures, but they had not had a chance to try it in practice before. This was new to them, as it had not been included in any of their other courses. It was one of the three parts of AAA that they found harder to imagine in real life. While they had often encountered authentication and authorization as users, accounting had remained abstract until now. The lab helped them clearly see how session data is collected, stored, and used, and made the concept much more concrete and understandable.

Step	Experienced Student (min)	New Student (min)
Step 1: Initial Configuration	20	27
Step 2: RADIUS Authentication	26	34
Step 3: Security Profiles	13	17
Step 4: Local Profiling	12	14
Step 5: ACL	15	19
Step 6: MAC Filtering	8	10
Step 7: Local Management Users	7	7
Step 8: Web Authentication	17	22
Step 9: RADIUS Accounting	31	37
Total Time	149	187

Both students finished the tasks without any issues and were capable of demonstrating correct configurations. All aspects of the lab were graded positively, and no major issues were reported.

5.2.3 Lab 3

During testing this lab, several issues were identified that affected the original design.

FlexConnect local authentication was part of the first version of the lab. The aim was to demonstrate how access points could authenticate users locally without relying on the controller. During the implementation task, however, it was discovered that the Wireless LAN Controller could not push its local user database to the APs. As a result, local authentication did not work as expected and the functionality could not be tested. This part of the lab was removed.

A similar issue occurred with the Local Split ACL feature. This ACL was intended to allow specific traffic, such as VPN connections, to bypass central control and go directly to local resources. However, in practice, the results were not consistent. The same configuration worked for some devices and services but failed for others. This inconsistency made it difficult to demonstrate the feature in a controlled lab environment. It is possible that the differences were caused by internal VPN restrictions, differences in how operating systems handle routing, or pre-configured security settings on test devices. Due to unpredictable behaviour, the Local Split ACL configuration was excluded from the lab.

Another issue was related to the Application Visibility and Control feature. The original intention was to classify more types of traffic, not just HTTP. However, it was discovered that most Internet traffic today uses HTTPS encryption. Because of this, the version of the AVC engine which we have is unable to identify the exact applications or content being accessed. Instead, it displayed all such traffic under a general "HTTPS" label. Although the configuration steps still worked and statistics were collected, the results were too general to fully demonstrate the potential of AVC in a classroom environment.

Here is a breakdown with step-by-step time stamps it took to complete each of the tasks.

Step	Experienced Student (min)	New Student (min)
Step 1: Initial Configuration	20	25
Step 2: LAGP	6	7
Step 3: Mobility Group Configuration	8	10
Step 4: FlexConnect VLAN Template Creation	7	12
Step 5: Creating a FlexConnect Group	10	16
Step 6: WLAN Configuration and Enabling Flex-Connect	7	11
Step 7: Enabling FlexConnect Mode on AP	6	9
Step 8: RADIUS Server	11	10
Step 9: Connecting to the Internet	5	5
Step 10: Testing the Wireless Connection	3	4
Step 11: FlexConnect ACL	12	20
Step 12: FlexConnect AVC	15	21
Step 13: FlexConnect Resiliency Scenarios N + N	20	32
Step 14: FlexConnect Resiliency Scenarios SSO	19	26
Step 15: FlexConnect resilience scenarios RADIUS	10	12
Total Time	159	220

Despite these issues, the rest of the lab steps were successfully completed by both students. The students were able to follow the steps, apply the configuration, verify the results, and demonstrate understanding of the material upon completion. These activities gave them a demonstration of how FlexConnect works.

5.3 Summary of Feedback

The validation of the lab exercises was carried out with two students whose profiles closely aligned with the intended target audience for the exercises. Although the measured completion times exceeded the originally planned 2.5-hour window, this was primarily due to additional theoretical explanations provided orally by the instructor throughout the process, as well as unforeseen technical difficulties that occurred due to this being the initial testing session. These technical issues were unrelated to the content of the labs and are unlikely to occur in regular classroom settings. As a result, they were considered insignificant in evaluating the effectiveness of the lab design. After taking these delays into account, the time data was slightly adjusted to better reflect a realistic classroom scenario.

It was observed that the student who already possessed relevant prior knowledge was able to successfully complete the lab tasks comfortably within the intended time frame of 2.5 hours. In contrast, the student who lacked sufficient prior experience

took more time to complete the exercises, largely because the instructor provided continuous verbal instruction and explanations aimed at fostering a deeper understanding of the theoretical concepts involved.

Overall, feedback from both participants was positive, indicating the labs were clear, practically useful, and educationally valuable. Participants also offered specific recommendations for improvement, which have since been incorporated into the final version. Implementing these suggested changes substantially improved the overall clarity, logical sequence, and educational effectiveness of the lab exercises, making them more engaging and accessible to future students.

6 Discussion and Conclusion

The goal of this work was to design and implement laboratory exercises that would cover the topics of the CCNP Enterprise Wireless Implementation certification. The lab exercises were designed to provide practice in configuring Cisco wireless networks with a focus on basic wireless configurations and access point modes, wireless security and FlexConnect.

During the thesis, related literature was analysed in order to develop a structured approach to the design of the lab exercises. The certification guide was used as a primary reference for selecting relevant topics, ensuring that the exercises addressed key aspects of wireless network implementation. This led to the creation of three laboratory exercises, each of them had objectives, network topology, step-by-step instructions, and methods for verifying completion of the exercises.

The exercises were validated through testing to confirm that the labs were possible to perform within the time frame given and with available equipment. Feedback was gathered and adjustments were made to improve the lab content. This helped reduce the time required to complete tasks and increased student engagement during the labs.

Despite the successful completion of three laboratory exercises, there were limitations due to time restrictions related to the thesis submission deadline. The goal was to create at least two lab exercises that were relevant to the CCNP Enterprise Wireless Implementation certification. The time available to develop additional lab exercises was not sufficient. The scope of the labs was limited to topics such as wireless network security and FlexConnect. While excluding areas such as quality of service, multicast configuration, location services, and device hardening. These topics are part of the CCNP certification and would have increased the educational value of the lab series. Future works should focus on expanding the number of laboratory exercises that cover these topics.

Another limitation was the use of AireOS. Currently, AireOS is used more in academic or legacy environments, and the rest of the industry is using IOS XE as the standard (Magrane, 2021). Future labs should incorporate IOS XE to align with industry practices and offer students more industry-relevant experience in configuring Cisco wireless. Modernizing the labs to include IOS XE would also open up opportunities to cover features that can't be implemented otherwise.

The validation of the laboratory exercises was done with a small group of participants due to time limitations. Although this testing provided insights into task clarity, the findings could be more complex. The work would benefit from expanding the testing phase to a larger group, or possibly adding the labs to existing courses, to provide more data. This extended testing would also allow for the identification of issues that may not have become clear in the initial testing period.

Integrating troubleshooting scenarios could also improve the value of the labs by requiring students to apply knowledge in a context where configurations are intentionally misconfigured or partially completed. This method would shift the

focus from following instructions to diagnosing and correcting network problems.

The objective was met by creating and validating three lab exercises. The exercises were based on the Cisco ENWLSI 300-430 exam requirements and and were structured to fit within the lab time.

7 References

- BENHAM, Dave, 2019. *Cisco Catalyst 9100 Series: 802.11ax APs to be excited about* [online]. [visited on 2025-05-08]. Available from: <https://www.wifireference.com/2019/04/29/cisco-catalyst-9100-series-80211ax-aps-to-be-excited-about/>.
- BUTLER, Brandon; JIROVSKY, Petr, 2024. *Enterprise WLAN Market Grows 5.8% From Second to Third Quarter of 2024, IDC Worldwide WLAN Tracker Shows* [online]. [visited on 2025-04-11]. Available from: <https://www.idc.com/getdoc.jsp?containerId=prUS52838624>.
- CISCO SYSTEMS, 2008. *Introduction to 802.11 technology* [online]. [visited on 2025-05-08]. Available from: https://www.cisco.com/web/TH/assets/docs/seminar/200903_Night_Academy_3_1_WLAN_Introduction.pdf.
- CISCO SYSTEMS, 2009. *December 2009 Product Guide* [online]. [visited on 2025-05-04]. Available from: https://www.cisco.com/c/dam/global/pt_pt/assets/docs/Cisco_Small_Business_Product_guide_Dec_2009.pdf.
- CISCO SYSTEMS, 2010. *Integrating Troubleshooting into the Network Maintenance Process* [online]. [visited on 2025-05-04]. Available from: <https://www.ciscopress.com/articles/article.asp?p=1578504&seqNum=3>.
- CISCO SYSTEMS, 2013. *Resetting the Controller to Default Settings* [online]. [visited on 2025-05-04]. Available from: https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-4/configuration/guides/consolidated/b_cg74_CONSOLIDATED/m_resetting_controller_to_default_settings.pdf.
- CISCO SYSTEMS, 2014a. *Cisco Catalyst 2960-S and 2960 Series Switches with LAN Lite Software Data Sheet* [online]. [visited on 2025-05-04]. Available from: https://www.cisco.com/c/en/us/products/collateral/switches/catalyst-2960-series-switches/product_data_sheet0900aecd806b0bd8.html.
- CISCO SYSTEMS, 2014b. *Configuring a Spectrum Expert Connection* [online]. [visited on 2025-05-08]. Available from: https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-6/configuration-guide/b_cg76/b_cg76_chapter_010000111.pdf.
- CISCO SYSTEMS, 2018. *Application Visibility and Control Feature Deployment Guide* [online]. [visited on 2025-05-08]. Available from: https://www.cisco.com/c/en/us/td/docs/wireless/controller/technotes/8-8/AVC_8point8_dg.html.
- CISCO SYSTEMS, 2019. *Data DTLS* [online]. [visited on 2025-05-04]. Available from: https://www.cisco.com/c/en/us/td/docs/wireless/controller/technotes/8-8/AVC_8point8_dg.html.

[//www.cisco.com/c/en/us/td/docs/wireless/controller/9800/config-guide/b_wl_16_10_cg/data-dtls.pdf](https://www.cisco.com/c/en/us/td/docs/wireless/controller/9800/config-guide/b_wl_16_10_cg/data-dtls.pdf).

CISCO SYSTEMS, 2021a. *Cisco 3504 Wireless Controller Data Sheet* [online]. [visited on 2025-05-04]. Available from: <https://www.cisco.com/c/en/us/products/collateral/wireless/3504-wireless-controller/datasheet-c78-738484.html>.

CISCO SYSTEMS, 2021b. *Cisco Wireless Controller Command Reference, Release 8.10* [online]. [visited on 2025-05-04]. Available from: <https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-10/cmd-ref/b-cr810.html>.

CISCO SYSTEMS, 2021c. *Cisco Wireless Controller Configuration Guide, Release 8.5* [online]. [visited on 2025-04-30]. Available from: https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-5/config-guide/b_cg85/radio_resource_management.html.

CISCO SYSTEMS, 2021d. *Cisco Wireless Controller Configuration Guide, Release 8.5* [online]. [visited on 2025-05-04]. Available from: https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-5/config-guide/b_cg85/flexconnect.html.

CISCO SYSTEMS, 2021e. *High Availability (SSO) Deployment Guide* [online]. [visited on 2025-05-08]. Available from: https://www.cisco.com/c/en/us/td/docs/wireless/controller/technotes/8-7/High_Availability_DG.html#pgfId-85015.

CISCO SYSTEMS, 2021f. *Managing Users* [online]. [visited on 2025-05-08]. Available from: https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-7/config-guide/b_cg87/m_managing_users.pdf.

CISCO SYSTEMS, 2022a. *Cisco Wireless Controller Configuration Guide, Release 8.10* [online]. [visited on 2025-04-30]. Available from: https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-10/config-guide/b_cg810.html.

CISCO SYSTEMS, 2022b. *Configure ACLs on Wireless LAN Controller Example* [online]. [visited on 2025-05-08]. Available from: <https://www.cisco.com/c/en/us/support/docs/wireless-mobility/wlan-security/71978-acl-wlc.html>.

CISCO SYSTEMS, 2023a. *Cisco Catalyst 9120 Access Point Deployment Guide* [online]. [visited on 2025-04-11]. Available from: <https://www.cisco.com/c/en/us/products/collateral/wireless/catalyst-9100ax-access-points/guide-c07-742311.html>.

- CISCO SYSTEMS, 2023b. *Cisco Catalyst 9120AX Series Access Points Data Sheet* [online]. [visited on 2025-05-04]. Available from: <https://www.cisco.com/c/en/us/products/collateral/wireless/catalyst-9120ax-series-access-points/datasheet-c78-742115.html>.
- CISCO SYSTEMS, 2023c. *Understand the AP Join Process with the Catalyst 9800 WLC* [online]. [visited on 2025-05-08]. Available from: <https://www.cisco.com/c/en/us/support/docs/wireless/catalyst-9120axe-access-point/221056-understand-the-ap-join-process-with-the.html>.
- CISCO SYSTEMS, 2023d. *Vendor Class Identifier DHCP Option 43, Option 60 and Option 241* [online]. [visited on 2025-05-04]. Available from: <https://community.cisco.com/t5/wireless-mobility-knowledge-base/vendor-class-identifier-dhcp-option-43-option-60-and-option-241/ta-p/5134724>.
- CISCO SYSTEMS, 2024a. *Cisco 4000 Family Integrated Services Router Data Sheet* [online]. [visited on 2025-05-04]. Available from: https://www.cisco.com/c/en/us/products/collateral/routers/4000-series-integrated-services-routers-isr/data_sheet-c78-732542.html.
- CISCO SYSTEMS, 2024b. *Configure AireOS Wireless LAN Controller and Lightweight Access Point* [online]. [visited on 2025-05-08]. Available from: <https://www.cisco.com/c/en/us/support/docs/wireless/4400-series-wireless-lan-controllers/69719-wlc-lwap-config.html>.
- CISCO SYSTEMS, 2024c. *Designing Cisco Enterprise Wireless Networks* [online]. [visited on 2025-03-04]. Available from: <https://www.cisco.com/site/us/en/learn/training-certifications/exams/enwlsd.html>.
- CISCO SYSTEMS, 2024d. *Implementing Cisco Enterprise Wireless Networks (300-430 ENWLSI)* [online]. [visited on 2025-04-22]. Available from: <https://www.cisco.com/site/us/en/learn/training-certifications/exams/enwlsi.html>.
- CISCO SYSTEMS, 2025a. *300-430 ENWLSI v1.1 Exam Topics* [online]. [visited on 2025-05-04]. Available from: <https://learningnetwork.cisco.com/s/enwlsi-exam-topics>.
- CISCO SYSTEMS, 2025b. *CCNA certification* [online]. [visited on 2025-05-04]. Available from: <https://www.cisco.com/site/us/en/learn/training-certifications/certifications/enterprise/ccna/index.html>.
- CISCO SYSTEMS, 2025c. *CCNP Enterprise certification* [online]. [visited on 2025-05-04]. Available from:

- <https://www.cisco.com/site/us/en/learn/training-certifications/certifications/enterprise/ccnp-enterprise/index.html>.
- CISCO SYSTEMS, 2025d. *Cisco Catalyst 9300 Series Switches Data Sheet* [online]. [visited on 2025-05-04]. Available from: <https://www.cisco.com/c/en/us/products/collateral/switches/catalyst-9300-series-switches/nb-06-cat9300-ser-data-sheet-cte-en.html>.
- CISCO SYSTEMS, 2025e. *Cisco Catalyst Center 2.3.7 Data Sheet* [online]. [visited on 2025-05-08]. Available from: <https://www.cisco.com/c/en/us/products/collateral/cloud-systems-management/dna-center/nb-06-dna-center-data-sheet-cte-en.html>.
- CISCO SYSTEMS, 2025f. *Cisco: AI Infrastructure, Secure Networking, and Software Solutions* [online]. [visited on 2025-05-04]. Available from: <https://www.cisco.com/>.
- CISCO SYSTEMS, 2025g. *Overview of Cisco ISE* [online]. [visited on 2025-05-08]. Available from: <https://www.cisco.com/site/us/en/products/security/identity-services-engine/index.html>.
- CISCO SYSTEMS, 2025h. *Software-Defined Access* [online]. [visited on 2025-05-08]. Available from: <https://www.cisco.com/site/us/en/solutions/networking/sdaccess/index.html>.
- CISCO SYSTEMS, 2025i. *What is a wireless access point?* [online]. [visited on 2025-05-08]. Available from: <https://www.cisco.com/site/us/en/learn/topics/small-business/what-is-an-access-point.html>.
- CISCO SYSTEMS, 2025j. *What is a wireless LAN?* [online]. [visited on 2025-05-08]. Available from: <https://www.cisco.com/site/us/en/learn/topics/security/what-is-a-wireless-lan.html>.
- CISCO SYSTEMS, 2025k. *What Is a WLAN Controller?* [online]. [visited on 2025-05-08]. Available from: <https://www.cisco.com/c/en/us/products/wireless/wireless-lan-controller/what-is-wlan-controller.html>.
- CISCO SYSTEMS, 2025l. *What Is IT (Information Technology)?* [online]. [visited on 2025-05-04]. Available from: <https://www.cisco.com/c/en/us/solutions/enterprise-networks/it-information-technology-explained.html>.
- COHERENT MARKET INSIGHTS, 2025. *Wireless Infrastructure Market Size and Share Analysis – Growth Trends and Forecasts (2025–2032)*

- [online]. [visited on 2025-04-21]. Available from:
<https://www.coherentmarketinsights.com/industry-reports/wireless-infrastructure-market>.
- EDGEWORTH, Brad; RIOS, Ramiro Garza; HUCABY, David;
GOOLEY, Jason, 2024. *CCNP and CCIE Enterprise Core ENCOR 350-401 Official Cert Guide*. 2nd ed. Cisco Press. ISBN 978-0-13-821676-4.
- ENAKYARHE, Omafume Matthew, 2011. *Wireless LAN Deployment: Single-Sign-On with Novell eDirectory Laholms kommun*. Sweden. Bachelor's Thesis. Halmstad University.
- FEE CTU, 2025. *NetAcad FEL ČVUT – Akademie síťové a informační bezpečnosti* [online]. [visited on 2025-04-21]. Available from:
<https://netacad.fel.cvut.cz/>.
- FIT BUT, 2025. *NetAcad at FIT – síťová akademie při FIT VUT* [online]. [visited on 2025-04-21]. Available from: <https://netacad.fit.vutbr.cz/>.
- FREERADIUS, 2025a. *FreeRADIUS* [online]. [visited on 2025-05-08]. Available from: <https://freeradius.org/>.
- FREERADIUS, 2025b. *The RADIUS Protocol* [online]. [visited on 2025-05-08]. Available from:
<https://networkradius.com/doc/current/introduction/RADIUS.html>.
- FREERADIUS PROJECT, 2018. *Getting Started* [online]. [visited on 2025-05-08]. Available from:
<https://wiki.freeradius.org/guide/Getting-Started#installing-the-server>.
- GOH, Jun Yu; CHUA, Darren Wei, 2024. A comparative analysis of wired vs. wireless data communication technologies. *International Journal of Electrical and Data Communication*. Vol. 5, no. 2, p. 12.
- HENRY, Jerome; HUCABY, David, 2024. *CCNP enterprise wireless design ENWLSD 300-425 and implementation ENWLSI 300-430 Official cert guide*. 2nd Edition. Indianapolis: Cisco Press. ISBN 978-0-13-824989-2.
- JAAKONSAARI, Mikko, 2019. *Implementation of Cisco FlexConnect*. Turku. Bachelor's Thesis. Turku University of Applied Sciences.
- KUMAR, Naveen, 2025. *Smartphone Usage Statistics 2025 (Worldwide Data)* [online]. [visited on 2025-04-22]. Available from:
<https://www.demandsage.com/smartphone-usage-statistics/>.
- LEMKE, Daniel, 2022. *The Different Types of Manuals in Technical Writing* [online]. [visited on 2025-05-04]. Available from:
https://library.lsbu.ac.uk/evaluating_resources/sources.

- LONDON SOUTH BANK UNIVERSITY LIBRARY, 2025. *Different types of sources – Evaluating Resources, how-to guide* [online]. [visited on 2025-05-04]. Available from: https://library.lsbu.ac.uk/evaluating_resources/sources.
- LUXES, n.d. *LED Technology* [online]. [visited on 2025-05-08]. Available from: <https://luxes.es/led-technology/>.
- MAGRANE, Byron, 2021. *After 15 years, Cisco Wireless AireOS Controllers are going away* [online]. [visited on 2025-05-08]. Available from: <https://blogs.cisco.com/networking/after-15-years-cisco-wireless-aireos-controllers-are-going-away>.
- MUNAWAR, Saima; TOOR, Saba Khalil; ASLAM, Muhammad; HAMID, Muhammad, 2018. Move to Smart Learning Environment: Exploratory Research of Challenges in Computer Laboratory and Design Intelligent Virtual Laboratory for eLearning Technology. *Eurasia Journal of Mathematics, Science and Technology Education* [online]. Vol. 14, no. 5, pp. 1645–1662 [visited on 2025-04-21]. Available from: <https://www.ejmste.com/download/move-to-smart-learning-environment-exploratory-research-of-challenges-in-computer-laboratory-and-5376.pdf>.
- NOVELL, INC., 2010. *Setting Up Single Sign-On (SSO)* [online]. [visited on 2025-05-08]. Available from: https://www.novell.com/documentation/windows_client/windows_client_admin/data/bxii3sl.html.
- NOVELL, INC., 2015. *Understanding NetIQ eDirectory* [online]. [visited on 2025-05-08]. Available from: https://www.netiq.com/documentation/edirectory-9/edir_admin/data/fbadjaeh.html.
- ODOM, Wendell; WILKINS, Sean, 2017. *CCNA Routing and Switching 200-125 Official Cert Guide Library and Pearson Ucertify Network Simulator Academic Edition Bundle*. 1st. Indianapolis, Indiana: Cisco Press. ISBN 978-1-58720-673-3.
- OLADELE, Oluwaseyi Kolawole, 2024. Advances in Wireless Network Technologies and Their Impact on Computer Science Applications [online], p. 14 [visited on 2025-04-21]. Available from: https://www.researchgate.net/publication/383907998_Advances_in_Wireless_Network_Technologies_and_Their_Impact_on_Computer_Science_Applications.
- PAHLAVAN, Kaveh; KRISHNAMURTHY, Prashant, 2021. Evolution and Impact of Wi-Fi Technology and Applications: A Historical Perspective. *International Journal of Wireless Information Networks* [online]. Vol. 28,

- no. 1, pp. 3–19 [visited on 2025-04-22]. ISSN 1077-2639. Available from: <https://link.springer.com/article/10.1007/s10776-020-00501-8>.
- RIESEN, Siswa A. N. van; GIJLERS, Hannie; ANJEWIERDEN, Anjo A.; JONG, Ton de, 2019. The Influence of Prior Knowledge on the Effectiveness of Guided Experiment Design. *Interactive Learning Environments* [online]. Vol. 27, no. 7, pp. 1–22 [visited on 2025-04-22]. ISSN 1049-4820. Available from: https://www.researchgate.net/publication/334021603_The_influence_of_prior_knowledge_on_the_effectiveness_of_guided_experiment_design.
- SOCHOR, Tomáš, 2018. *Cisco CCNA Wireless* [online]. 1st ed. Mendel University in Brno [visited on 2025-05-03]. ISBN 978-80-7509-609-8. Available from: <https://emendelu.publi.cz/book/248-cisco-ccna-wireless>.
- TECHTARGET, 2022. *5 most valuable wireless network certifications* [online]. [visited on 2025-04-21]. Available from: <https://www.techtarget.com/searchnetworking/tip/Most-valuable-wireless-network-certifications>.
- VARGA, Michael, 2022. *Příprava laboratorní úlohy SD-access s využitím Cisco DNA Center*. Brno. Bachelor's Thesis. Mendel University in Brno.
- ZIPPIA, 2025. *Job outlook for wireless engineers in the United States* [online]. [visited on 2025-04-21]. Available from: <https://www.zippia.com/wireless-engineer-jobs/trends>.